Benchmarking Report

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1. Benchmarking report

2. Algorithms

The following algorithms are considered in this benchmarking report.

2.1 L_BFGS_B

Limited-memory BFGS algorithm implemented in the SciPy library

2.2 SLSQP

Sequential Least-Squares Quadratic Programming (SLSQP) implemented in the SciPy library

3. Problem configurations

The benchmarking reference problems are listed below.

- Rastrigin
- Rosenbrock
- Rosenbrock 5D

4. Results

The results of each group of algorithms configurations on each group of problem configurations.

- L-BFGS-B_configurations
- SLSQP

5. Problems

5.1 Rastrigin

5.1.1 Description

No description available.

Optimal feasible objective value: 0.

5.1.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

5.2 Rosenbrock 5D

5.2.1 Description

No description available.

Optimal feasible objective value: 0.

5.2.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

5.3 Rosenbrock

5.3.1 Description

No description available.

Optimal feasible objective value: 0.

5.3.2 Target values

- 0.0001 (feasible)
- 1e-05 (feasible)
- 1e-06 (feasible)
- 0.0 (feasible)

6. Results

6.1 L-BFGS-B configurations

- 2D_problems
- 5D_problems

6.2 SLSQP

- 2D_problems
- 5D_problems

6.3 L BFGS B configurations

6.3.1 2D problems

The results of the group of algorithms configurations "L-BFGS-B configurations" on the group of problems "2D problems".

The algorithms configurations

- L-BFGS-B with 2 Hessian corrections
- L-BFGS-B with 20 Hessian corrections

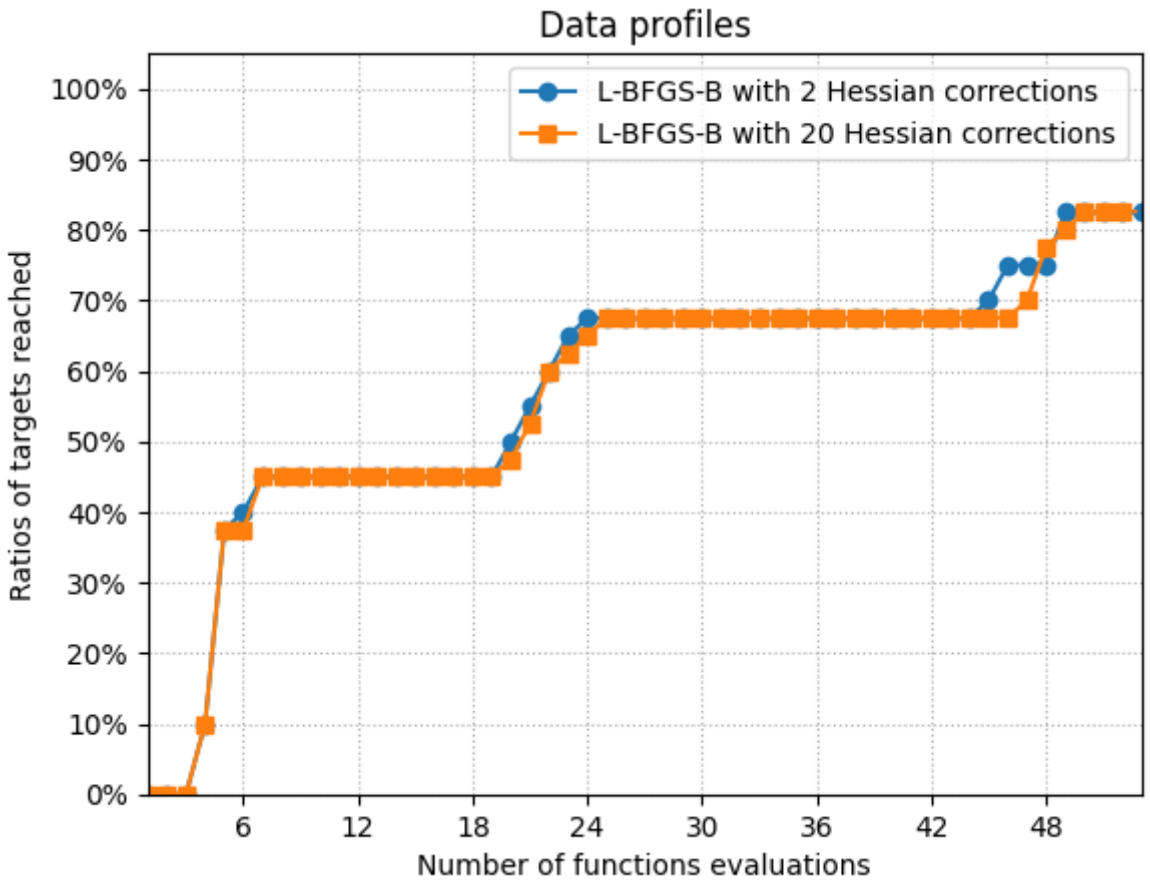
The problems

- Rastrigin
- Rosenbrock

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "2D problems" are represented in the following data profile.



The data profiles for group "2D problems".

RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

- Rastrigin
- Rosenbrock

6.3.2 5D problems

The results of the group of algorithms configurations "L-BFGS-B configurations" on the group of problems "5D problems".

The algorithms configurations

- L-BFGS-B with 2 Hessian corrections
- L-BFGS-B with 20 Hessian corrections

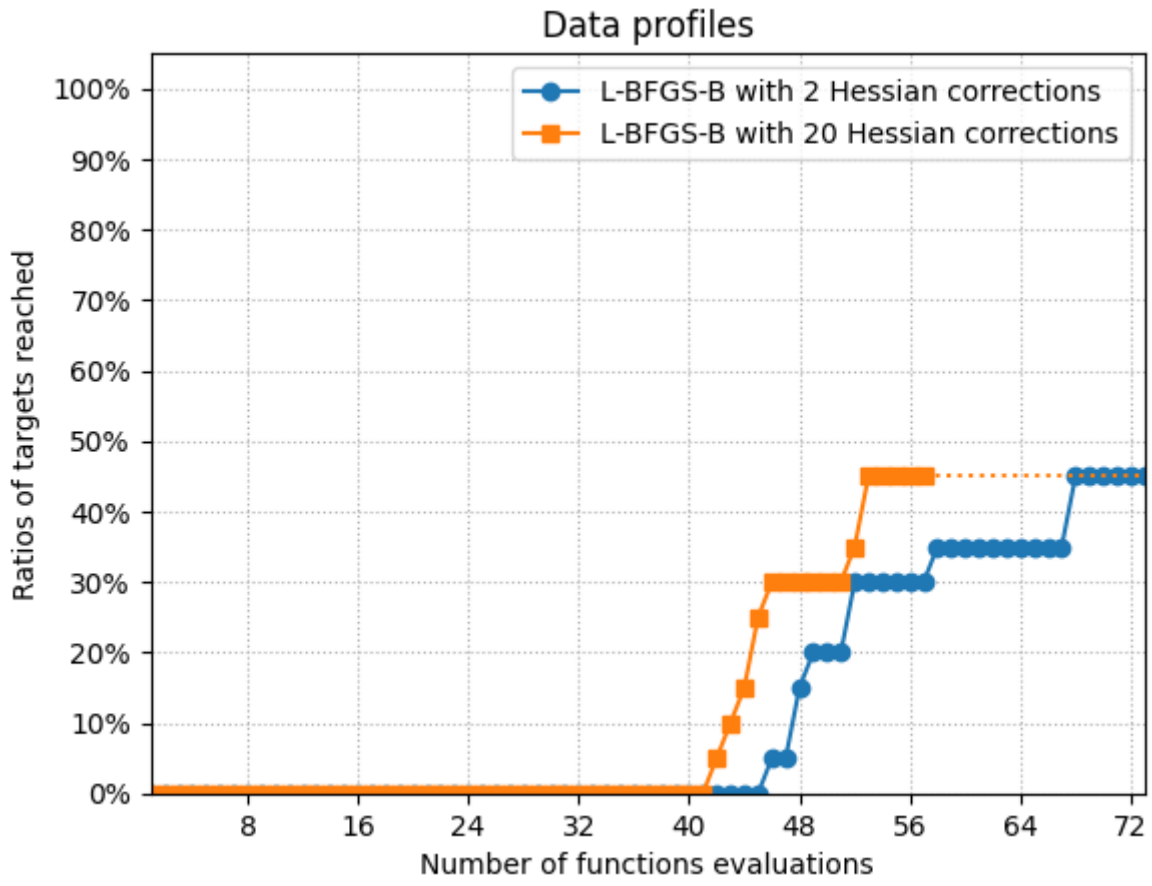
The problems

- Rosenbrock 5D

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "5D problems" are represented in the following data profile.



The data profiles for group "5D problems".

RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

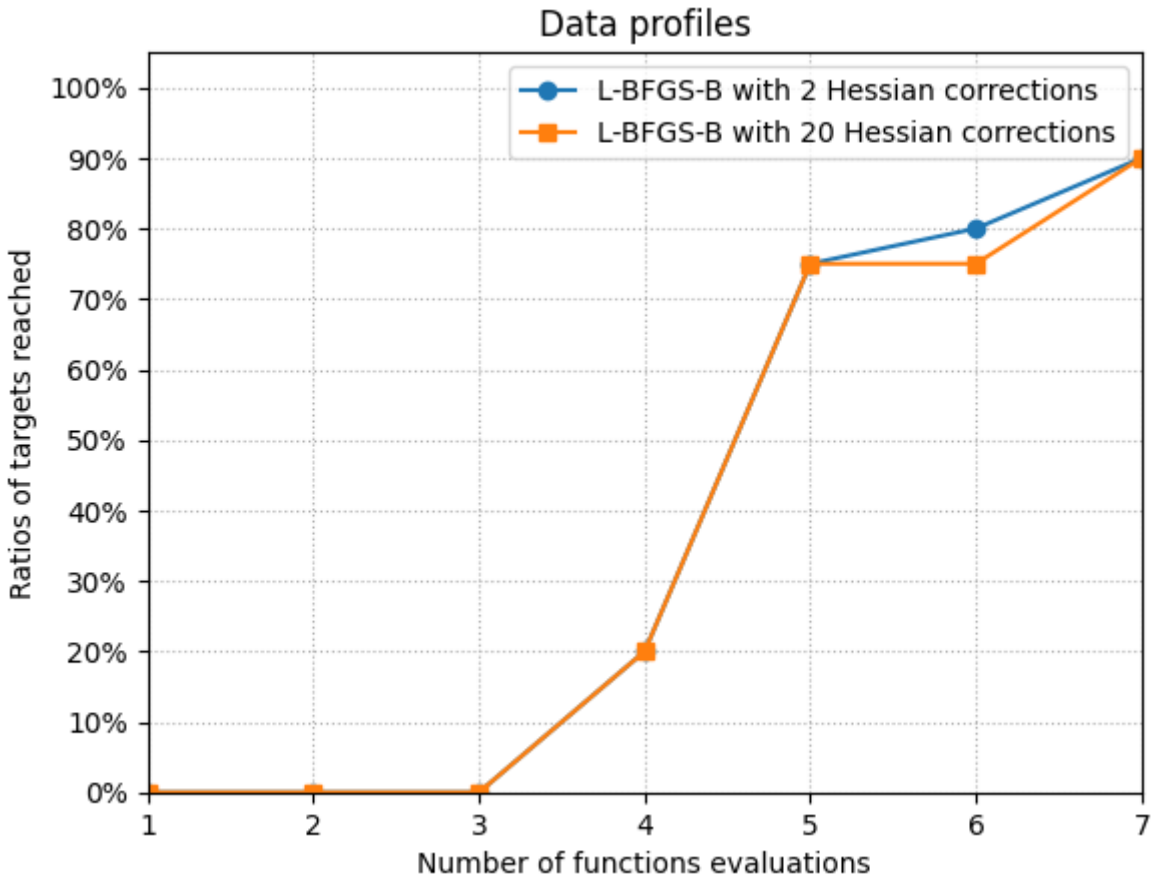
- Rosenbrock_5D

6.3.3 2D problems

Rastrigin

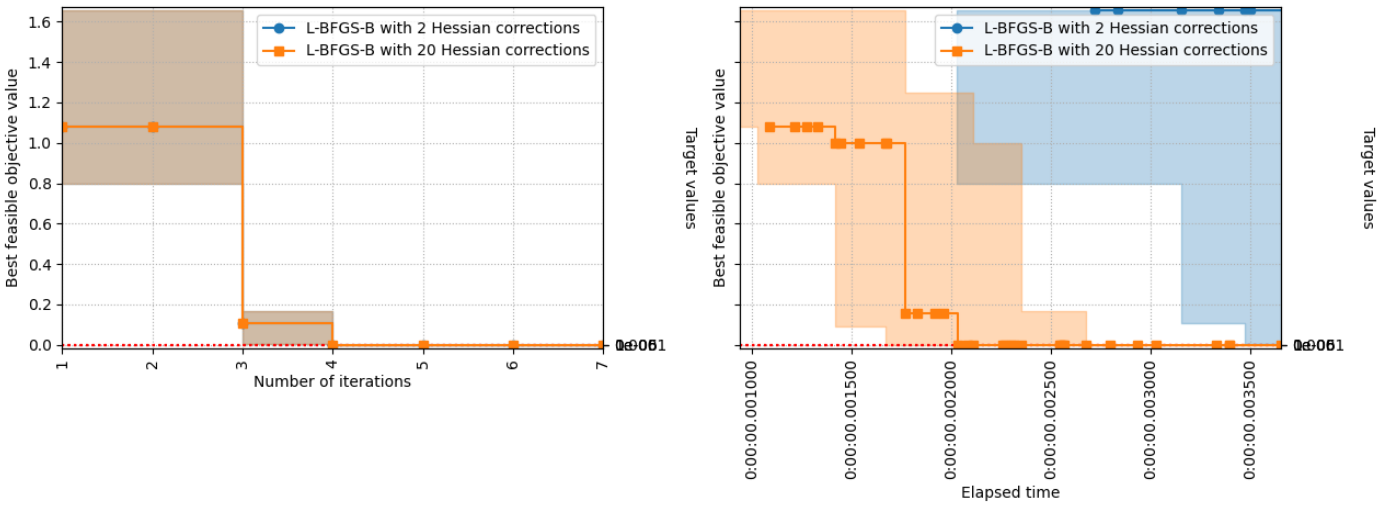
The results of the group of algorithms configurations 'L-BFGS-B configurations' for problem configuration 'Rastrigin'.

DATA PROFILES

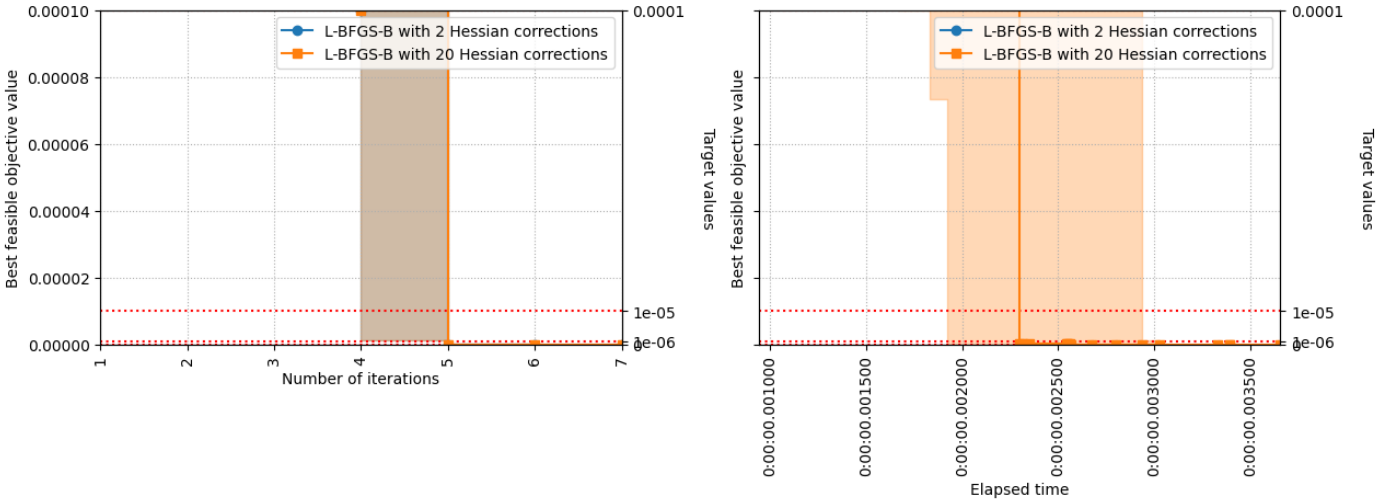


The data profiles of group L-BFGS-B configurations for problem configuration Rastrigin.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group L-BFGS-B configurations for problem configuration Rastrigin.

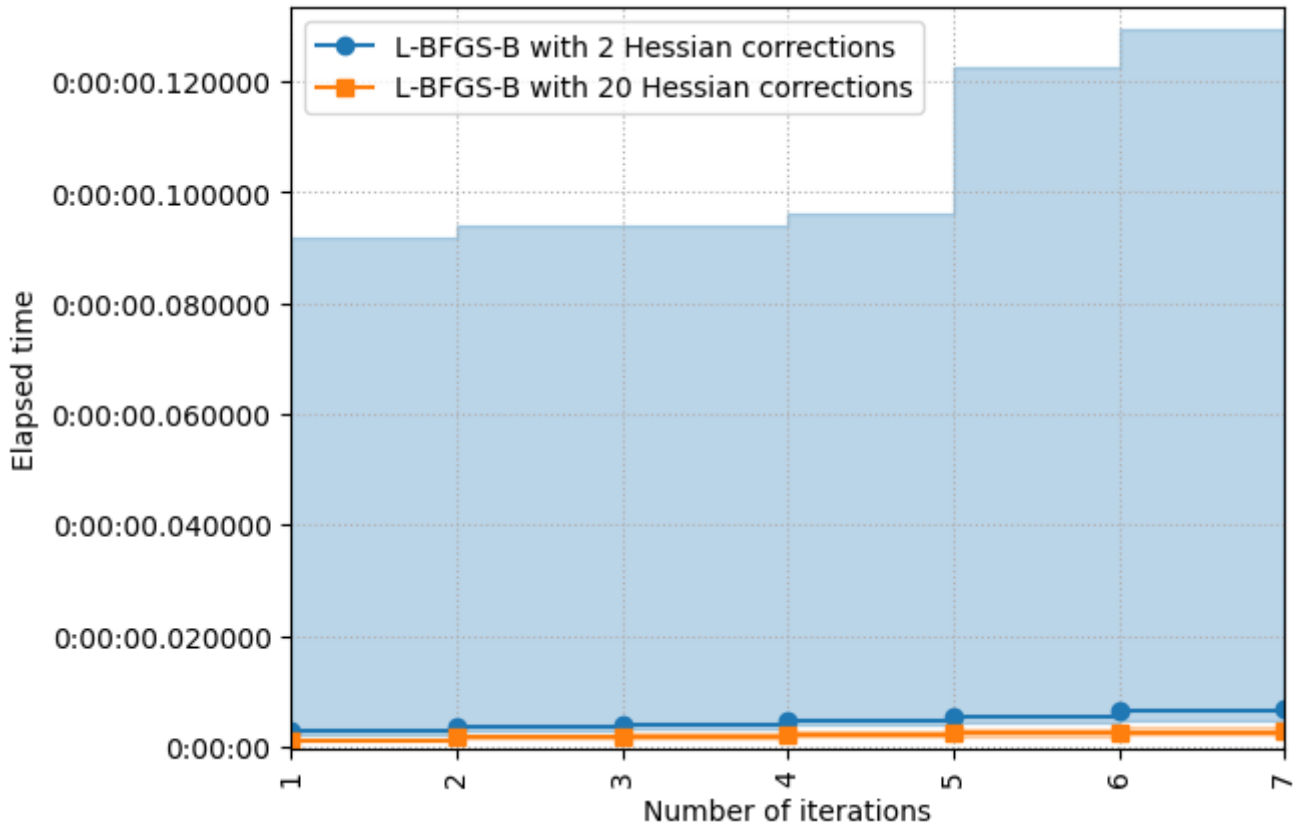


The best feasible objective value of group L-BFGS-B configurations for problem configuration Rastrigin.

	minimum	25th centile	median	75th centile	maximum
L-BFGS-B with 2 Hessian corrections	0	0	0	3.55271e-15	5.32907e-15
L-BFGS-B with 20 Hessian corrections	0	0	0	1.77636e-15	5.32907e-15

The final best feasible objective value of group L-BFGS-B configurations for problem configuration Rastrigin.

EXECUTION TIME



The execution time of group L-BFGS-B configurations for problem configuration Rastrigin.

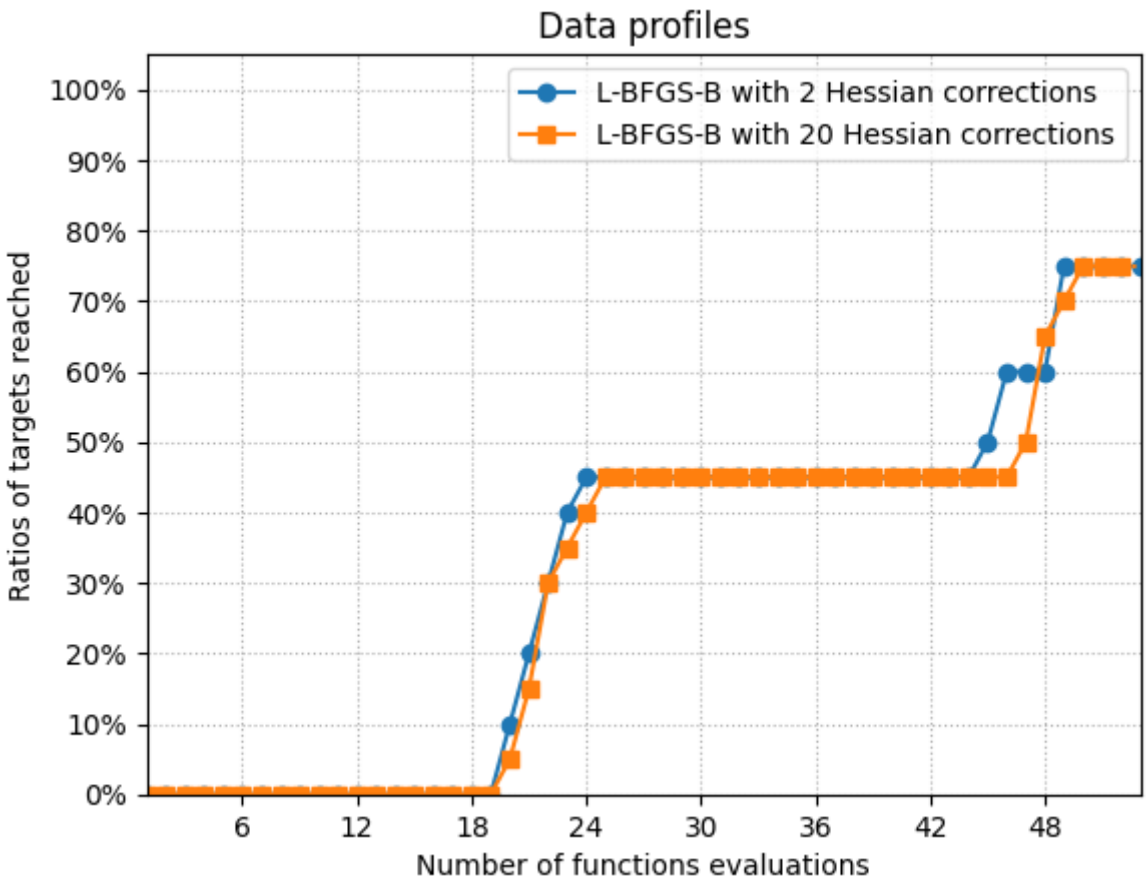
RESULTS FOR EACH ALGORITHM CONFIGURATION

- L-BFGS-B_with_2_Hessian_corrections
- L-BFGS-B_with_20_Hessian_corrections

Rosenbrock

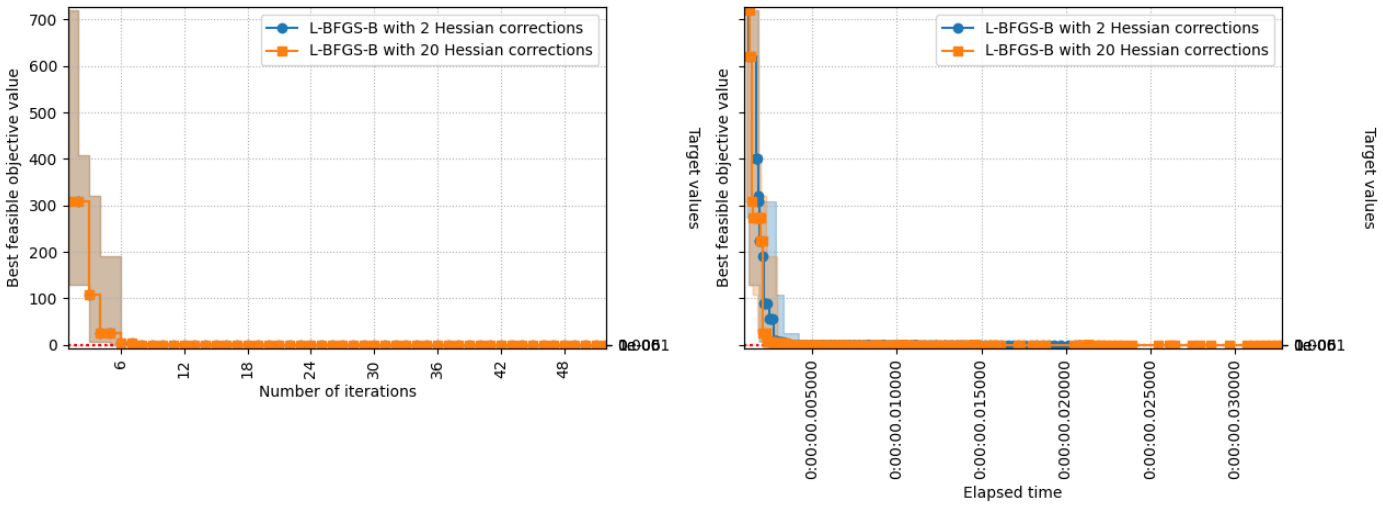
The results of the group of algorithms configurations 'L-BFGS-B configurations' for problem configuration 'Rosenbrock'.

DATA PROFILES

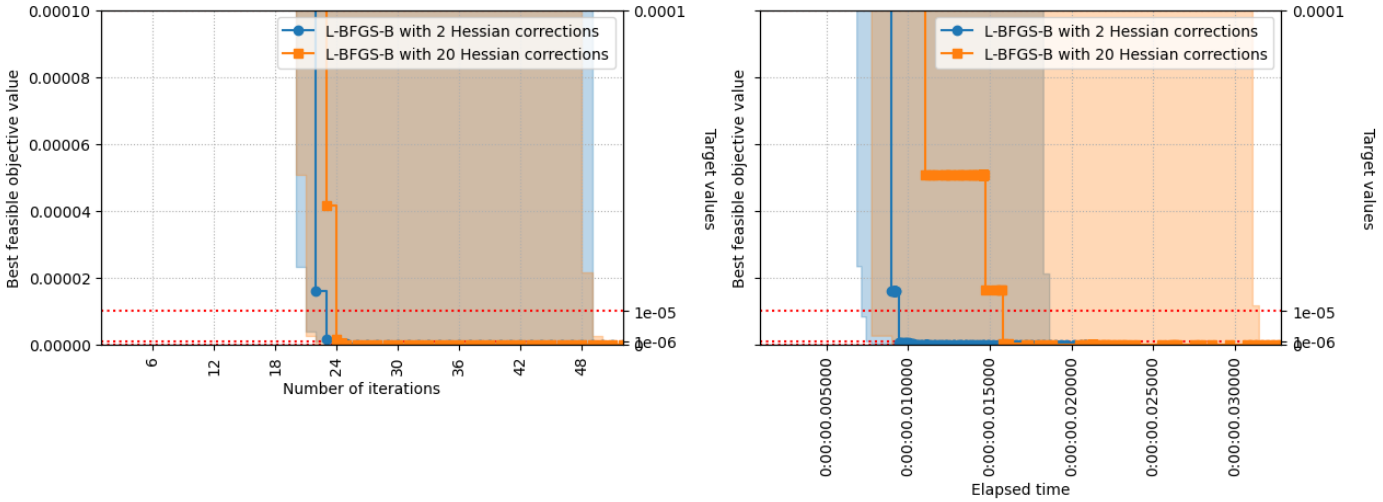


The data profiles of group L-BFGS-B configurations for problem configuration Rosenbrock.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock.

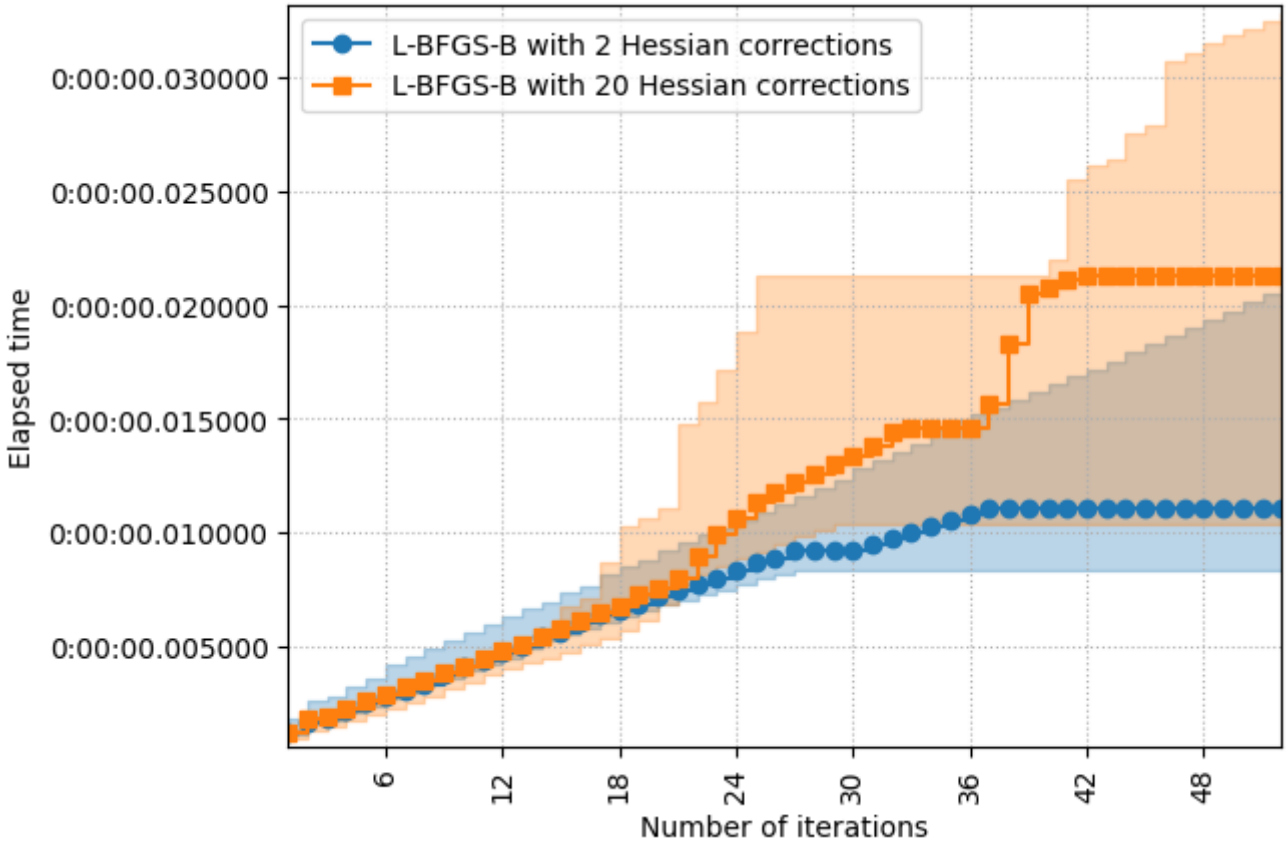


The best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock.

	minimum	25th centile	median	75th centile	maximum
L-BFGS-B with 2 Hessian corrections	1.74144e-20	1.19048e-17	1.13779e-15	3.35761e-15	4.68664e-13
L-BFGS-B with 20 Hessian corrections	3.40838e-20	2.33575e-19	1.4195e-18	1.76694e-18	6.79141e-14

The final best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock.

EXECUTION TIME



The execution time of group L-BFGS-B configurations for problem configuration Rosenbrock.

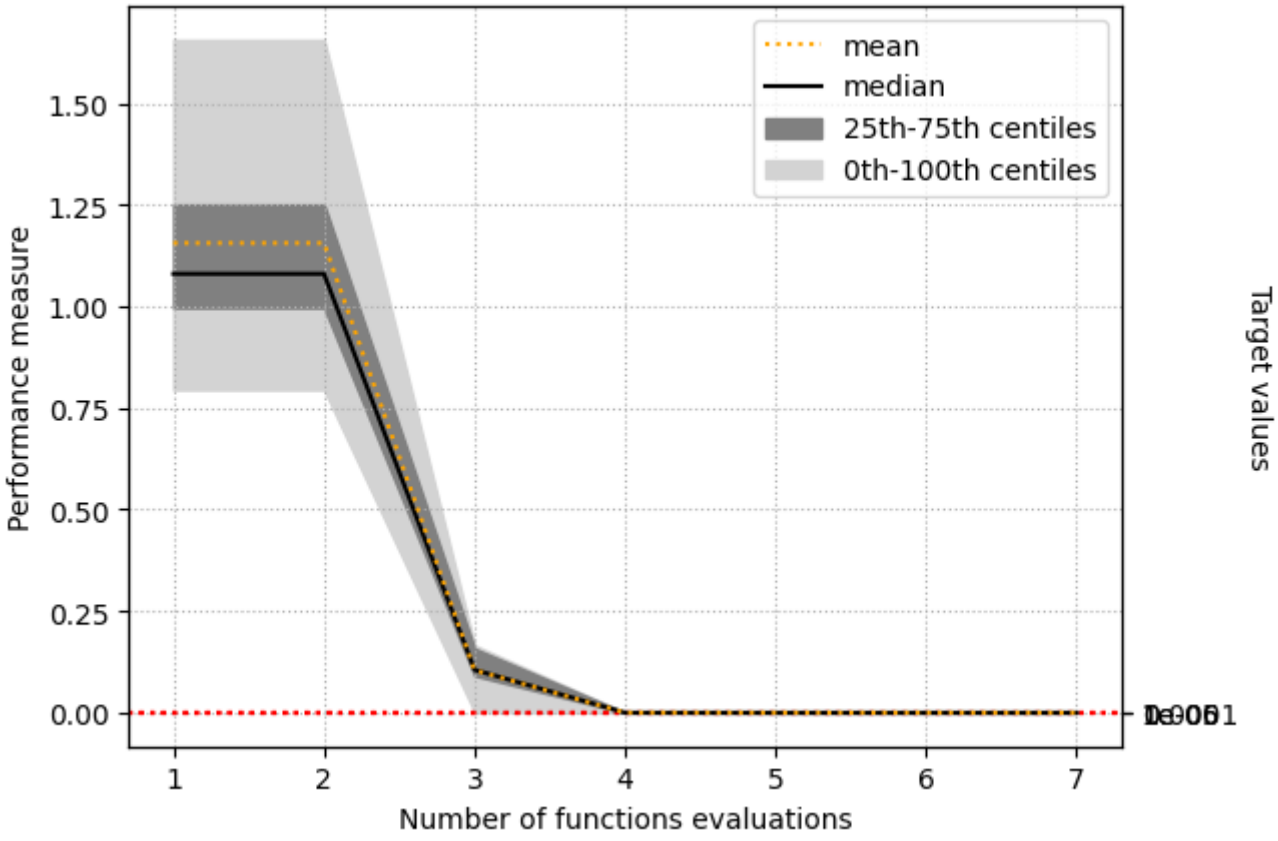
RESULTS FOR EACH ALGORITHM CONFIGURATION

- L-BFGS-B_with_2_Hessian_corrections
- L-BFGS-B_with_20_Hessian_corrections

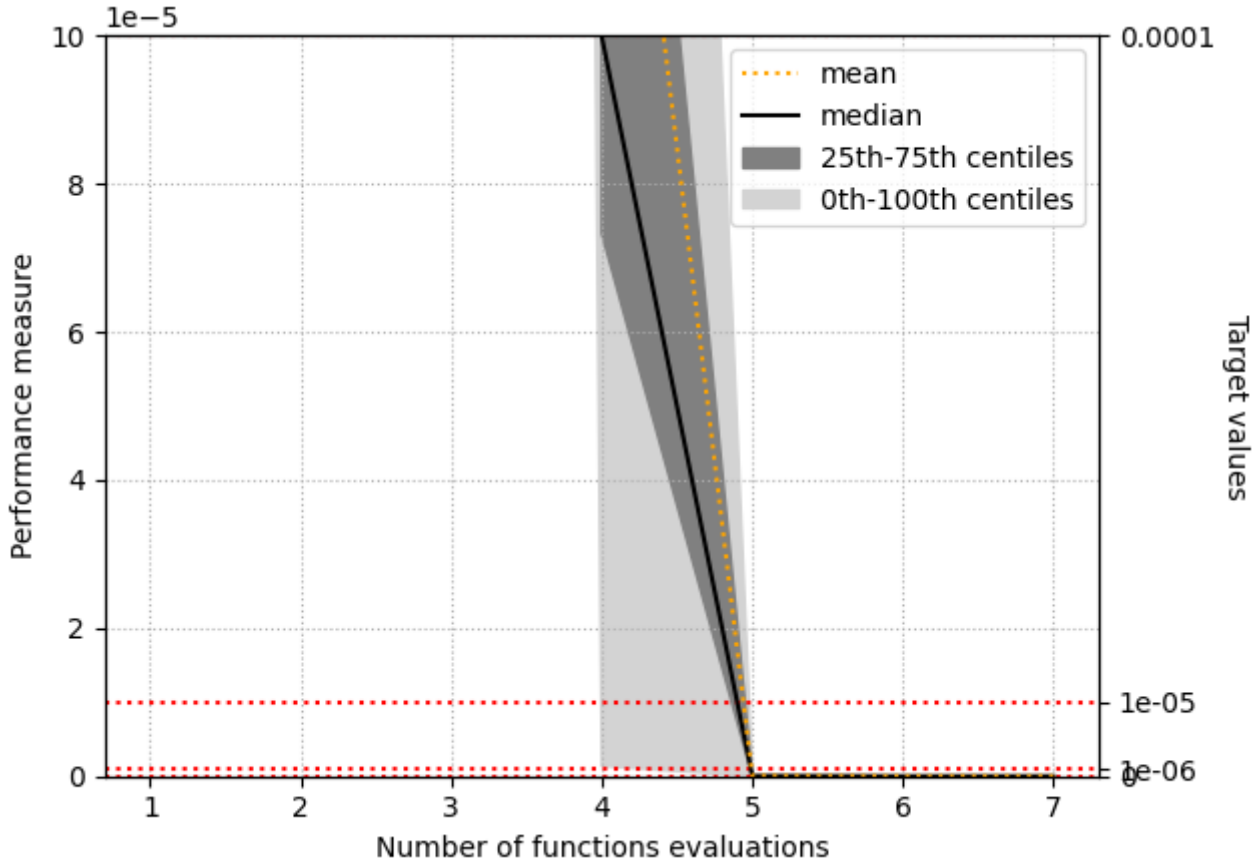
Rastrigin

L-BFGS-B WITH 20 HESSIAN CORRECTIONS ON RASTRIGIN

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rastrigin'.



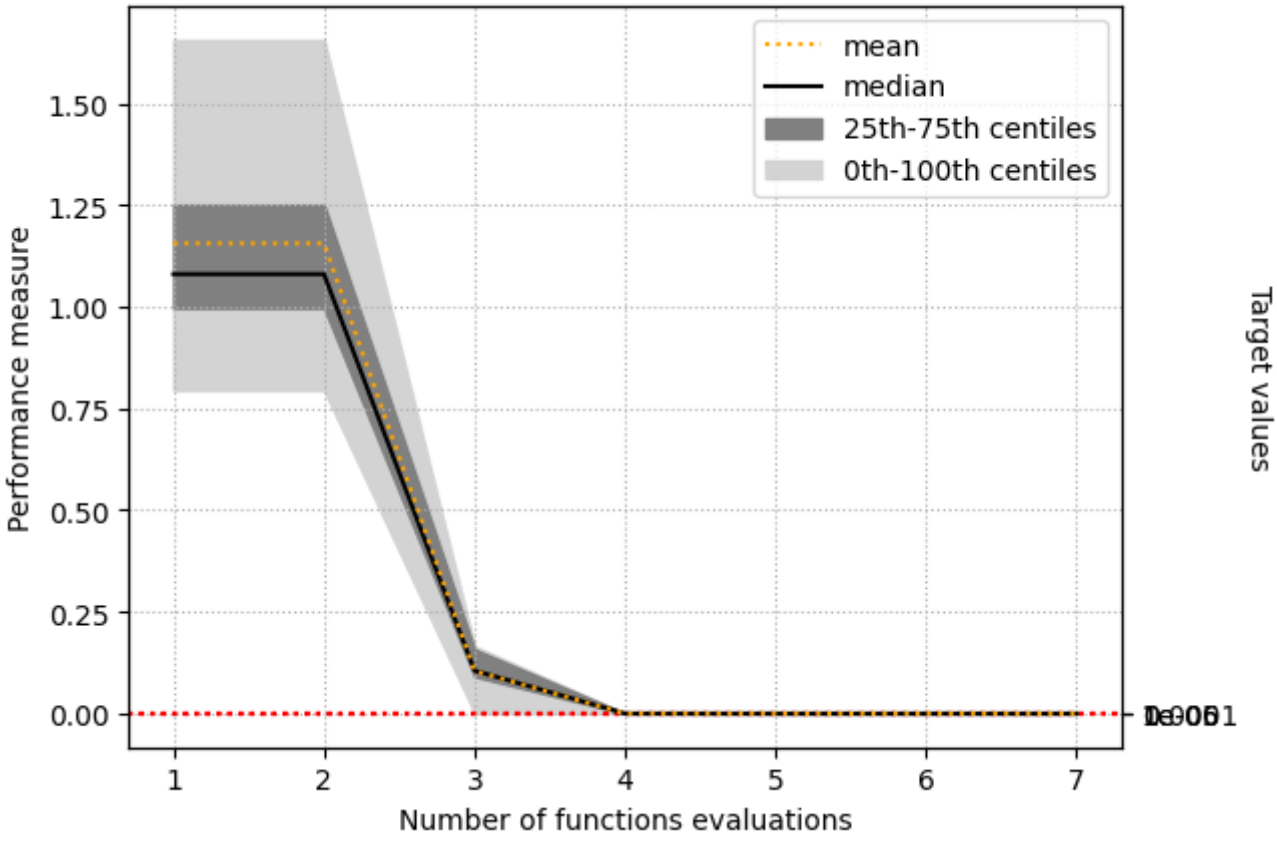
The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rastrigin'.

	L-BFGS-B with 20 Hessian corrections
maximum	5.32907e-15
75th centile	1.77636e-15
median	0
25th centile	0
minimum	0

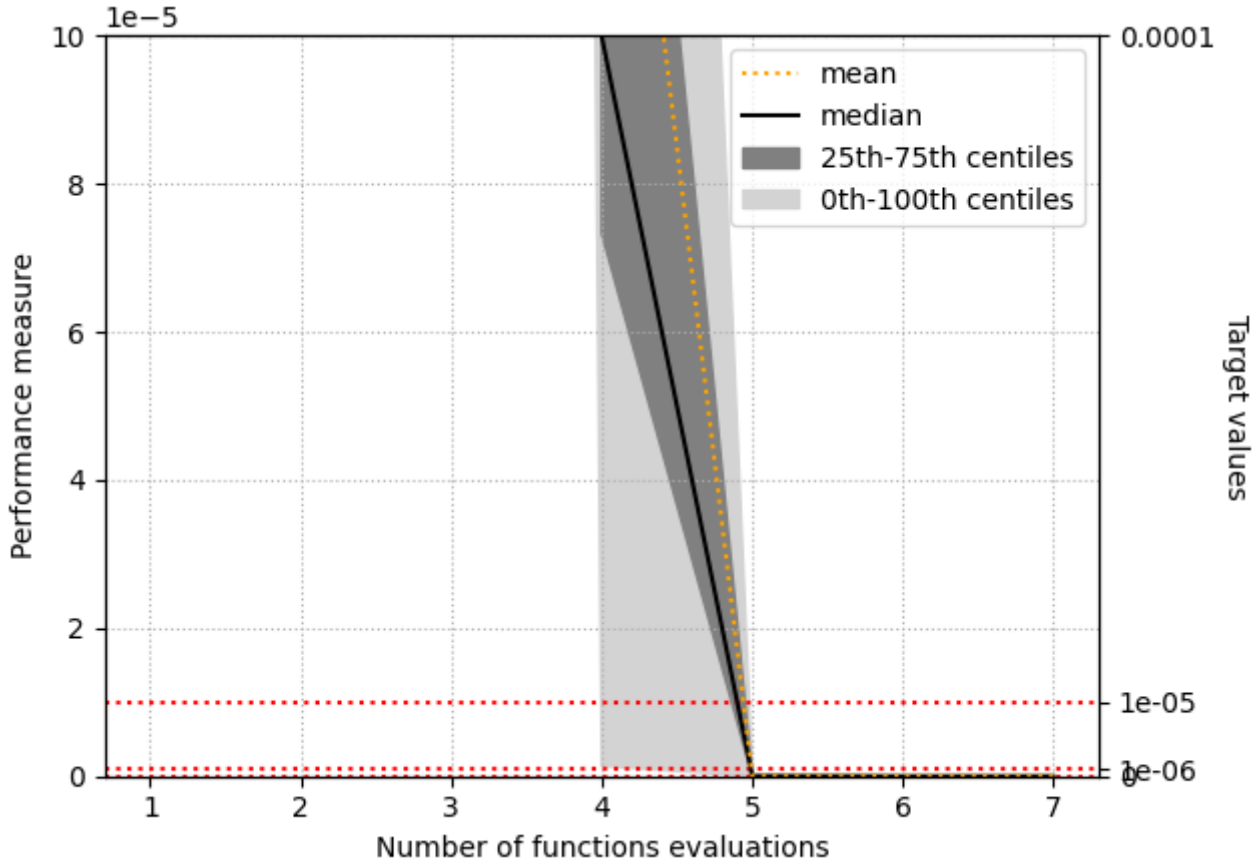
The final feasible performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem Rastrigin.

L-BFGS-B WITH 2 HESSIAN CORRECTIONS ON RASTRIGIN

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rastrigin'.



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rastrigin'.

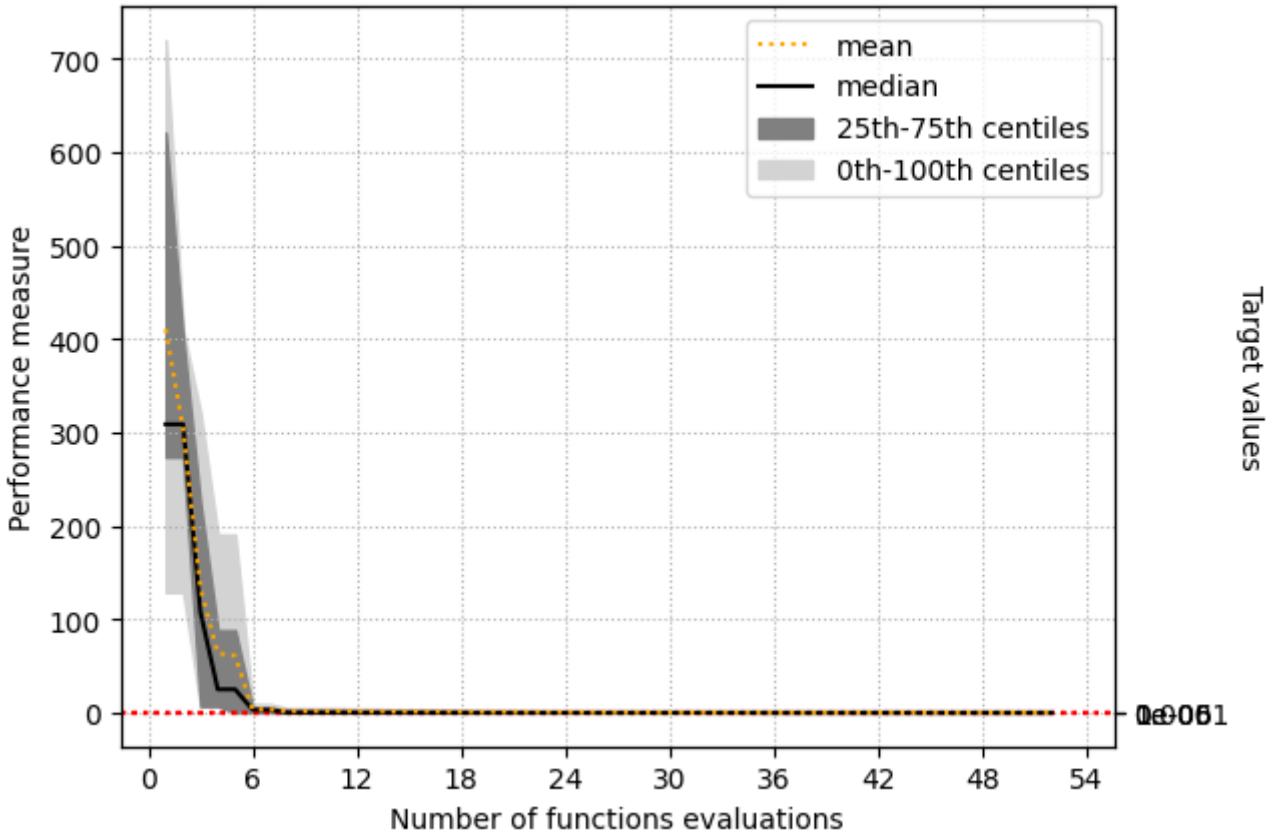
	L-BFGS-B with 2 Hessian corrections
maximum	5.32907e-15
75th centile	3.55271e-15
median	0
25th centile	0
minimum	0

The final feasible performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem Rastrigin.

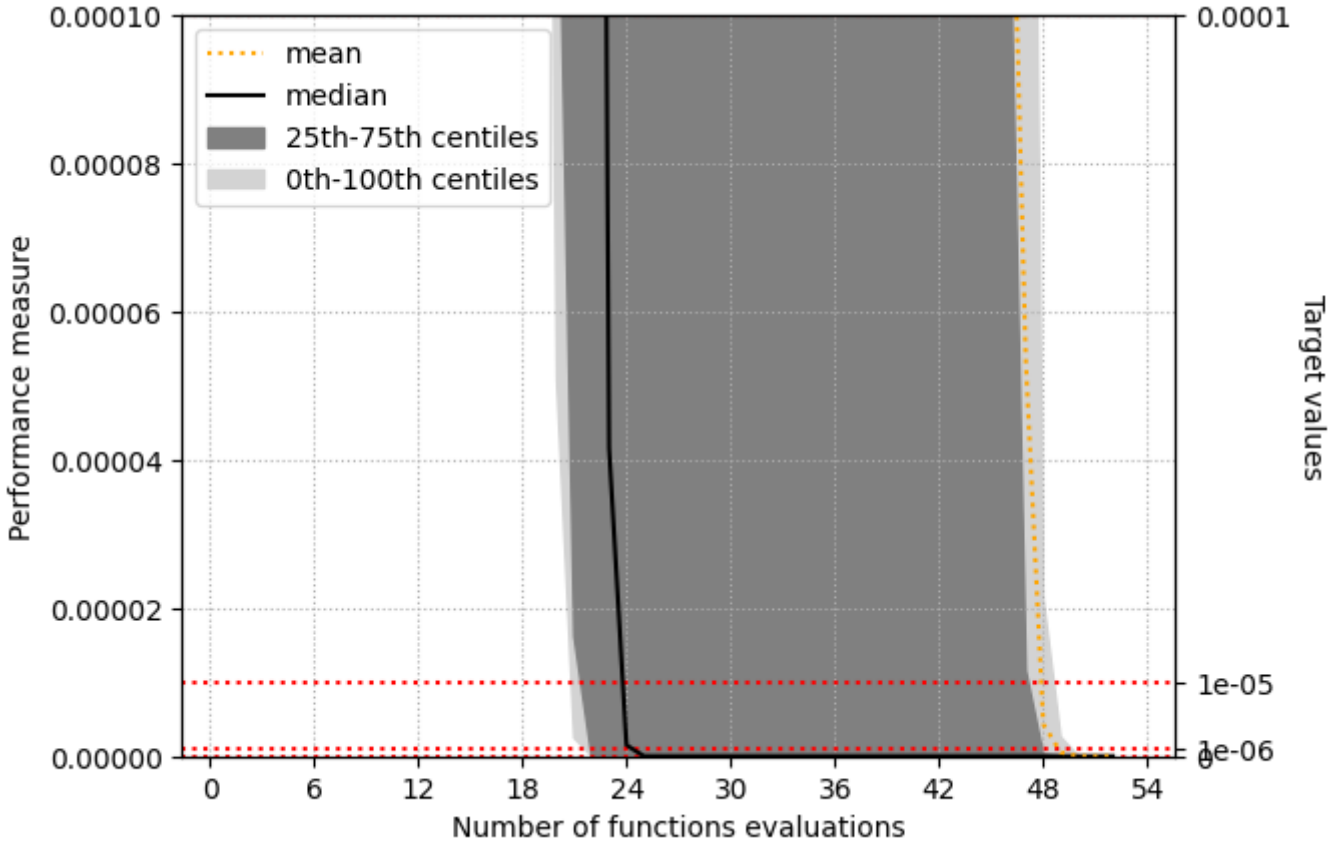
Rosenbrock

L-BFGS-B WITH 20 HESSIAN CORRECTIONS ON ROSENBROCK

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rosenbrock'.



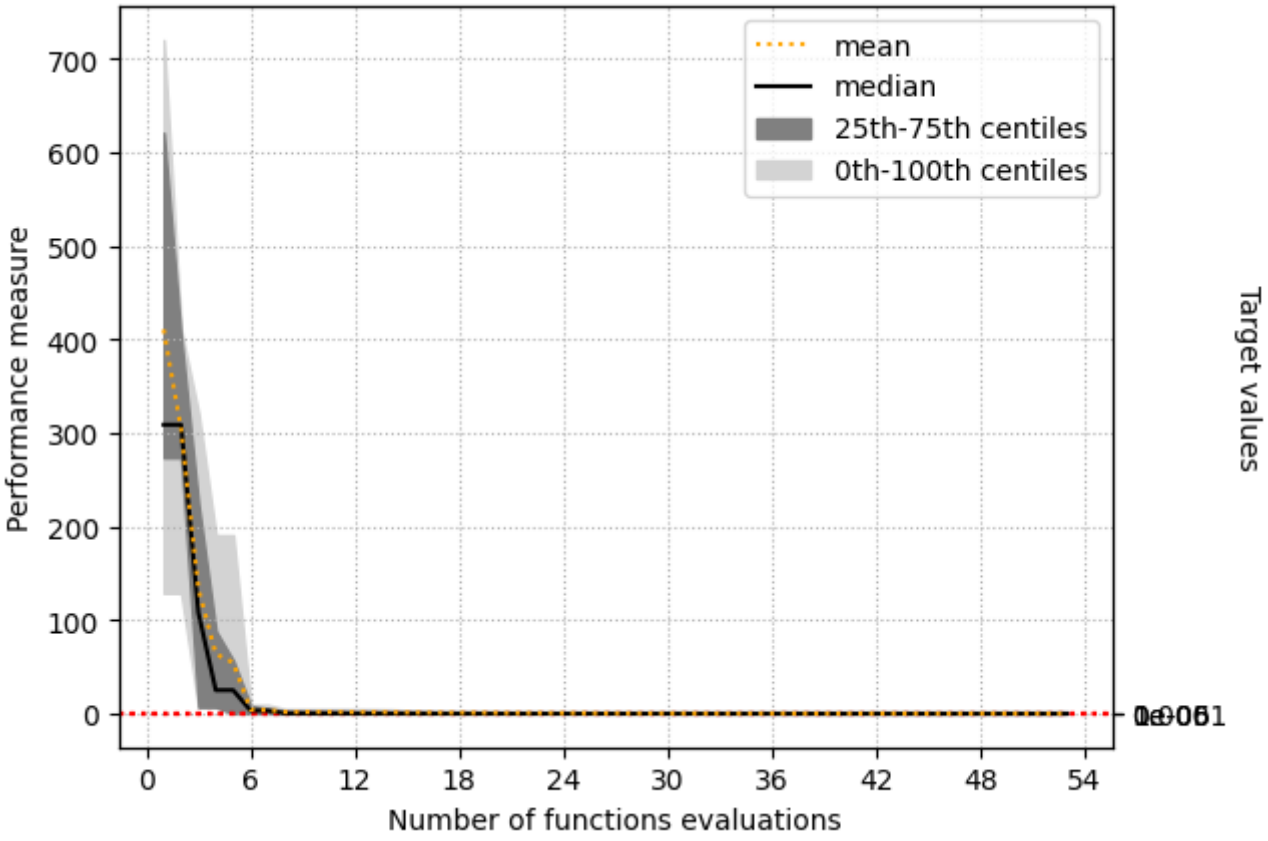
The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rosenbrock'.

	L-BFGS-B with 20 Hessian corrections
maximum	6.79141e-14
75th centile	1.76694e-18
median	1.4195e-18
25th centile	2.33575e-19
minimum	3.40838e-20

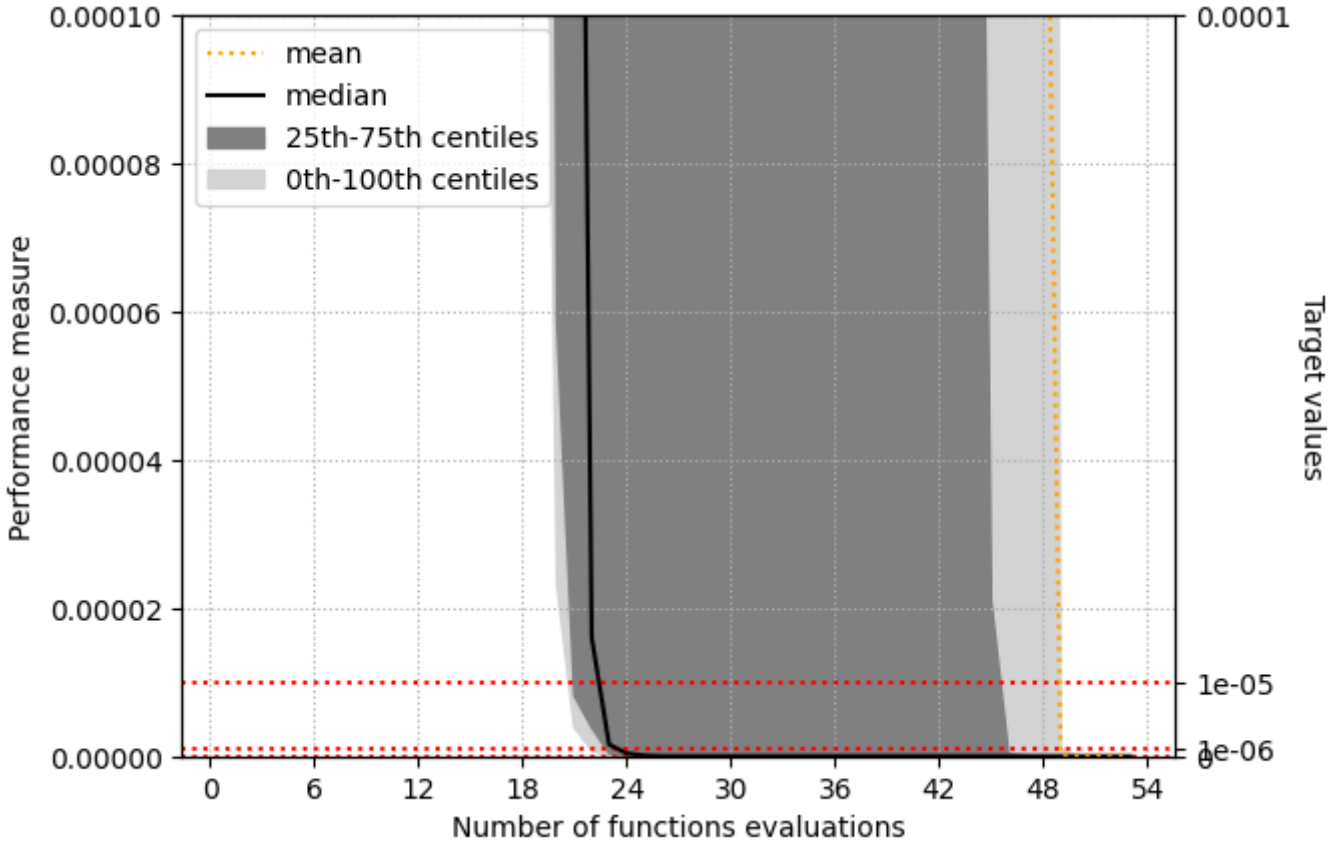
The final feasible performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem Rosenbrock.

L-BFGS-B WITH 2 HESSIAN CORRECTIONS ON ROSENBROCK

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rosenbrock'.



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rosenbrock'.

	L-BFGS-B with 2 Hessian corrections
maximum	4.68664e-13
75th centile	3.35761e-15
median	1.13779e-15
25th centile	1.19048e-17
minimum	1.74144e-20

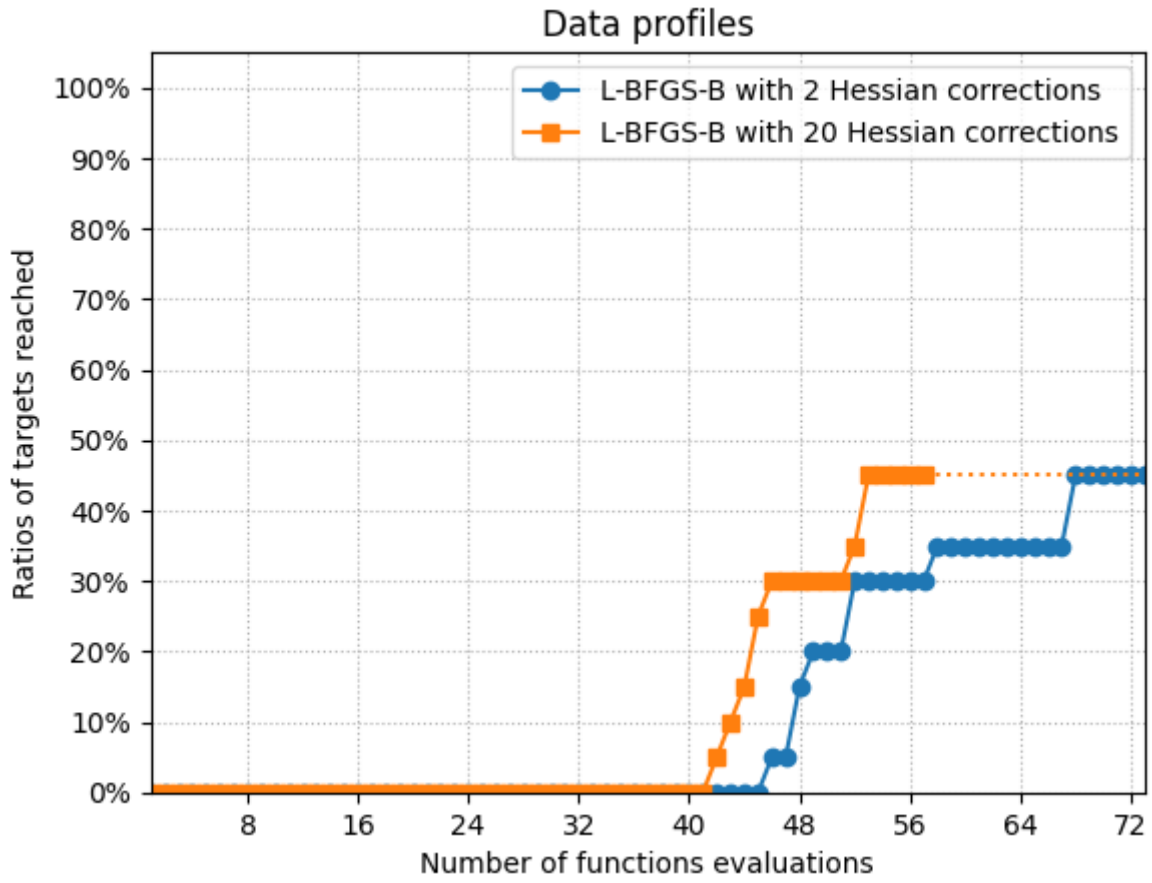
The final feasible performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem Rosenbrock.

6.3.4 5D problems

Rosenbrock 5D

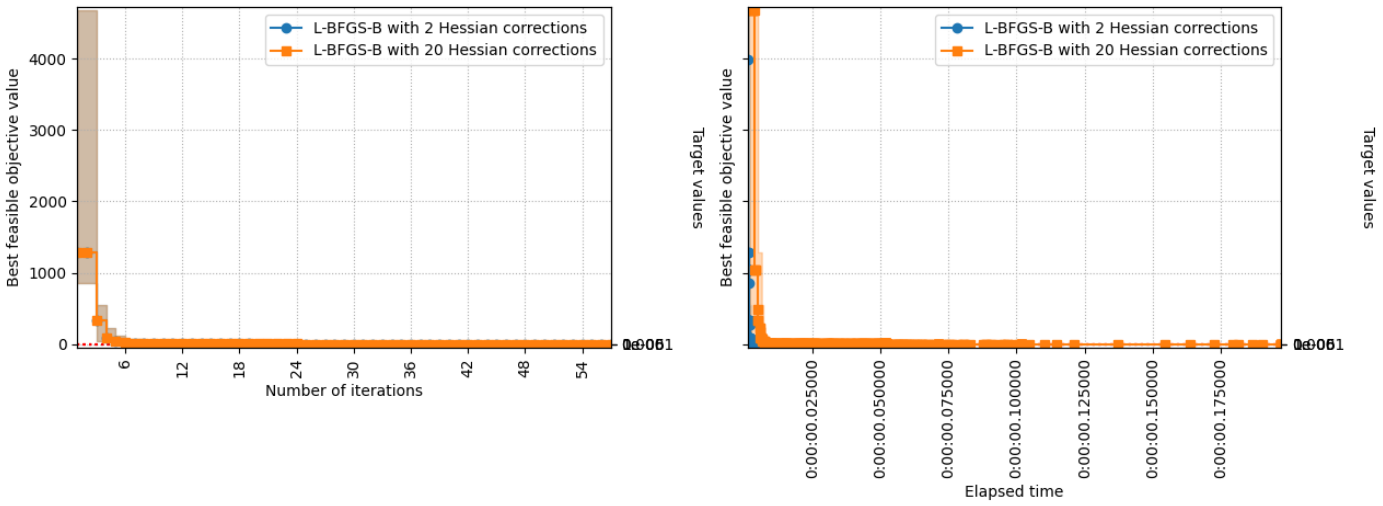
The results of the group of algorithms configurations 'L-BFGS-B configurations' for problem configuration 'Rosenbrock 5D'.

DATA PROFILES

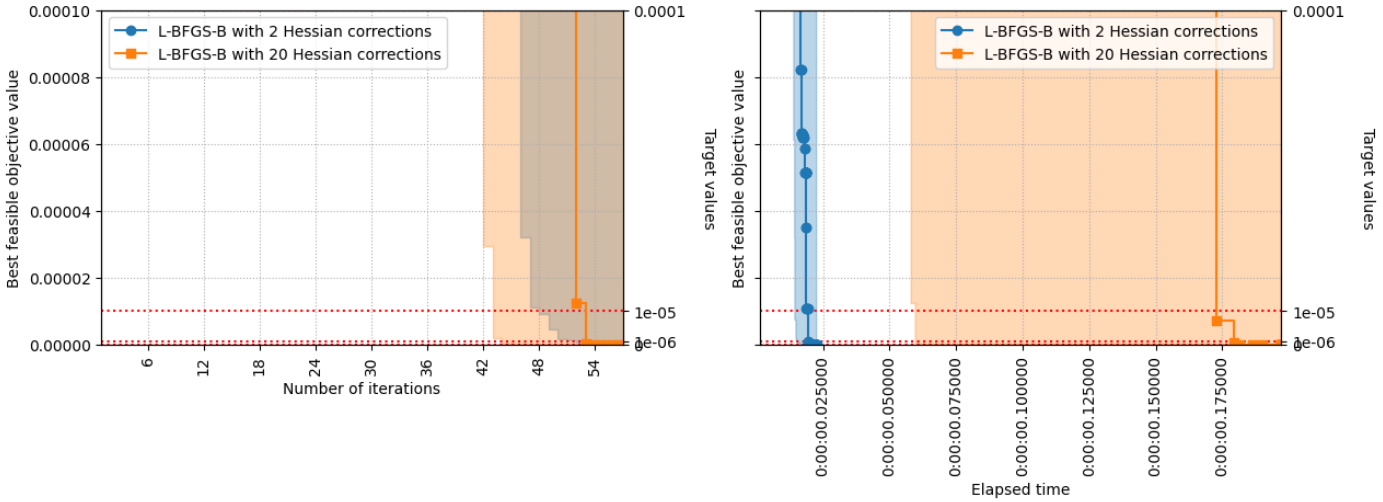


The data profiles of group L-BFGS-B configurations for problem configuration Rosenbrock 5D.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock 5D.

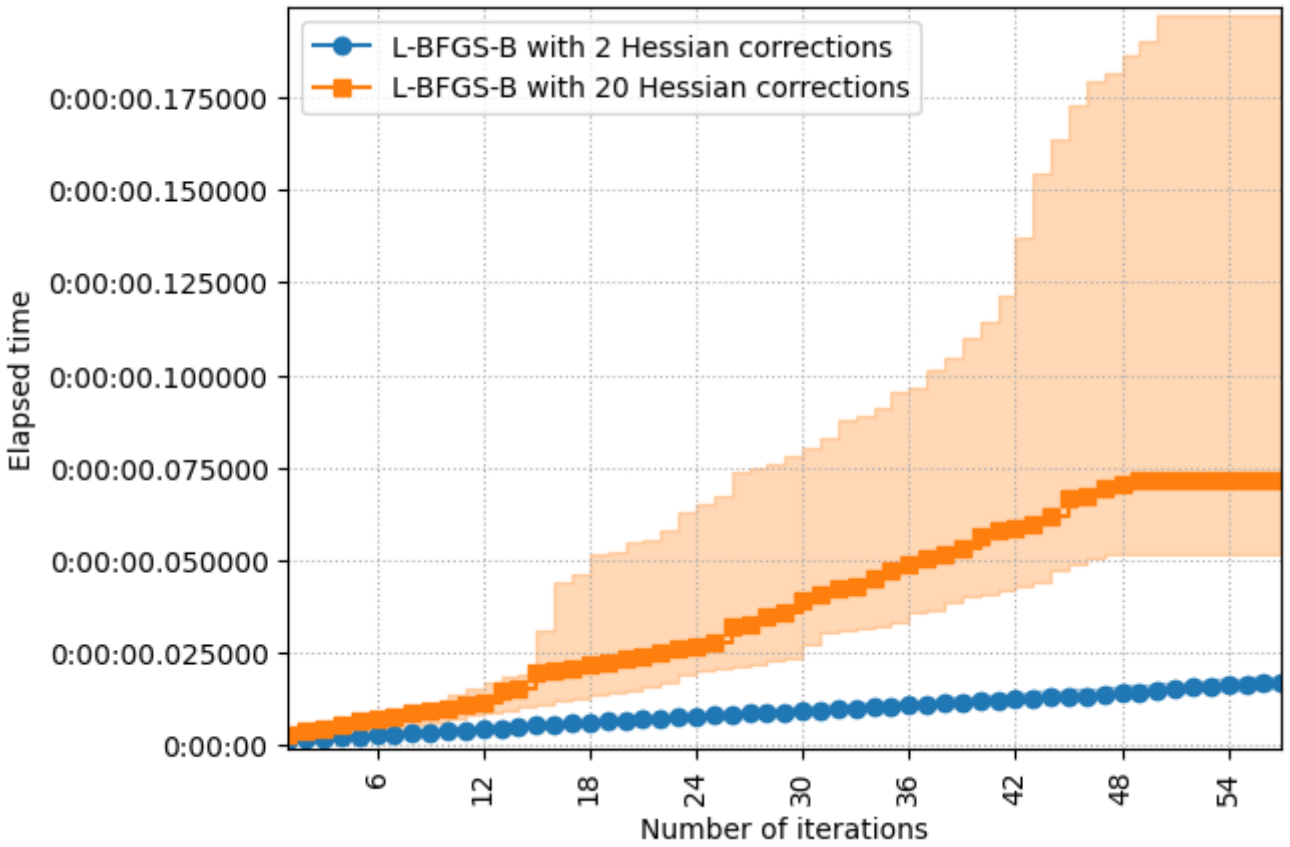


The best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock 5D.

	minimum	25th centile	median	75th centile	maximum
L-BFGS-B with 2 Hessian corrections	2.30683e-11	1.2383e-10	1.5986e-10	3.93084	3.93084
L-BFGS-B with 20 Hessian corrections	1.06233e-18	1.70659e-17	5.67236e-15	3.93084	3.93084

The final best feasible objective value of group L-BFGS-B configurations for problem configuration Rosenbrock 5D.

EXECUTION TIME



The execution time of group L-BFGS-B configurations for problem configuration Rosenbrock 5D.

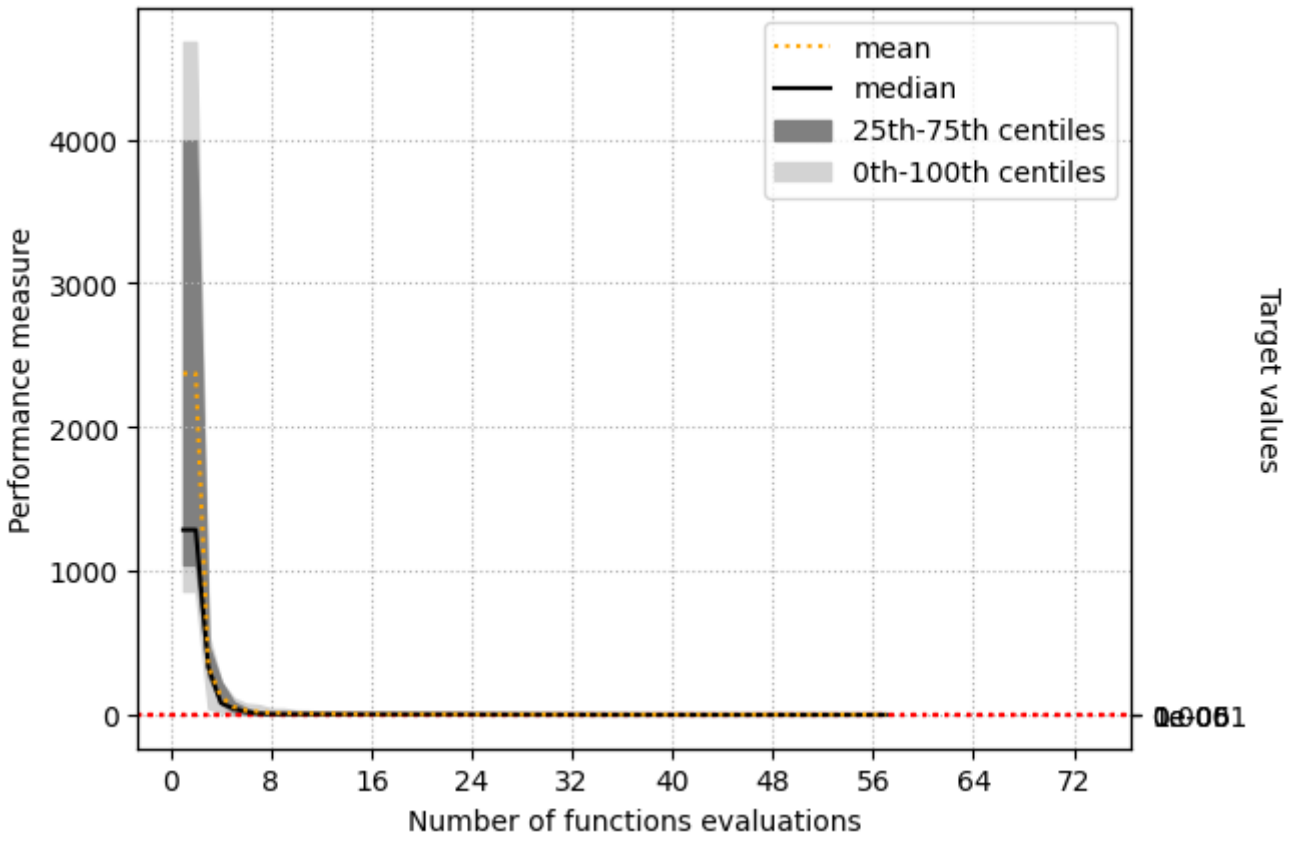
RESULTS FOR EACH ALGORITHM CONFIGURATION

- L-BFGS-B_with_2_Hessian_corrections
- L-BFGS-B_with_20_Hessian_corrections

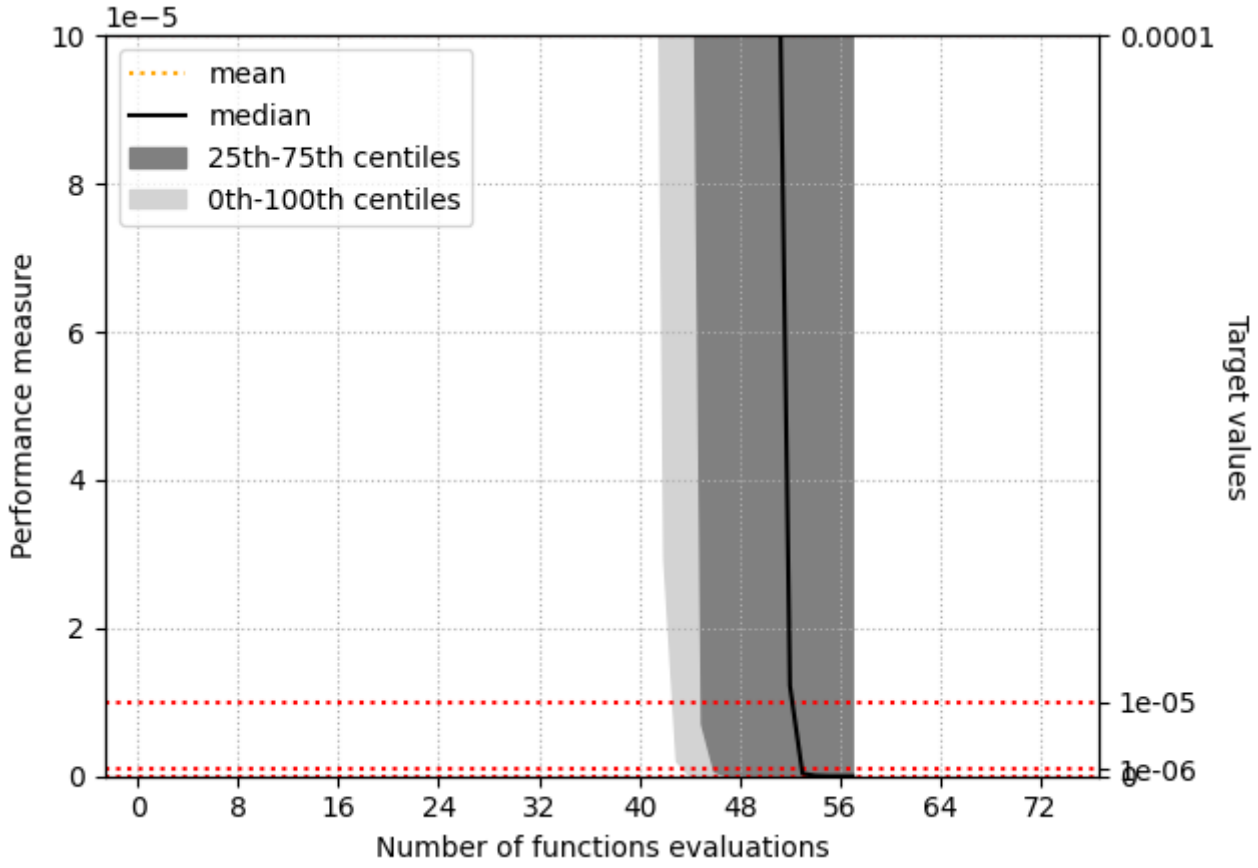
Rosenbrock 5D

L-BFGS-B WITH 20 HESSIAN CORRECTIONS ON ROSENBROCK 5D

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rosenbrock 5D'.



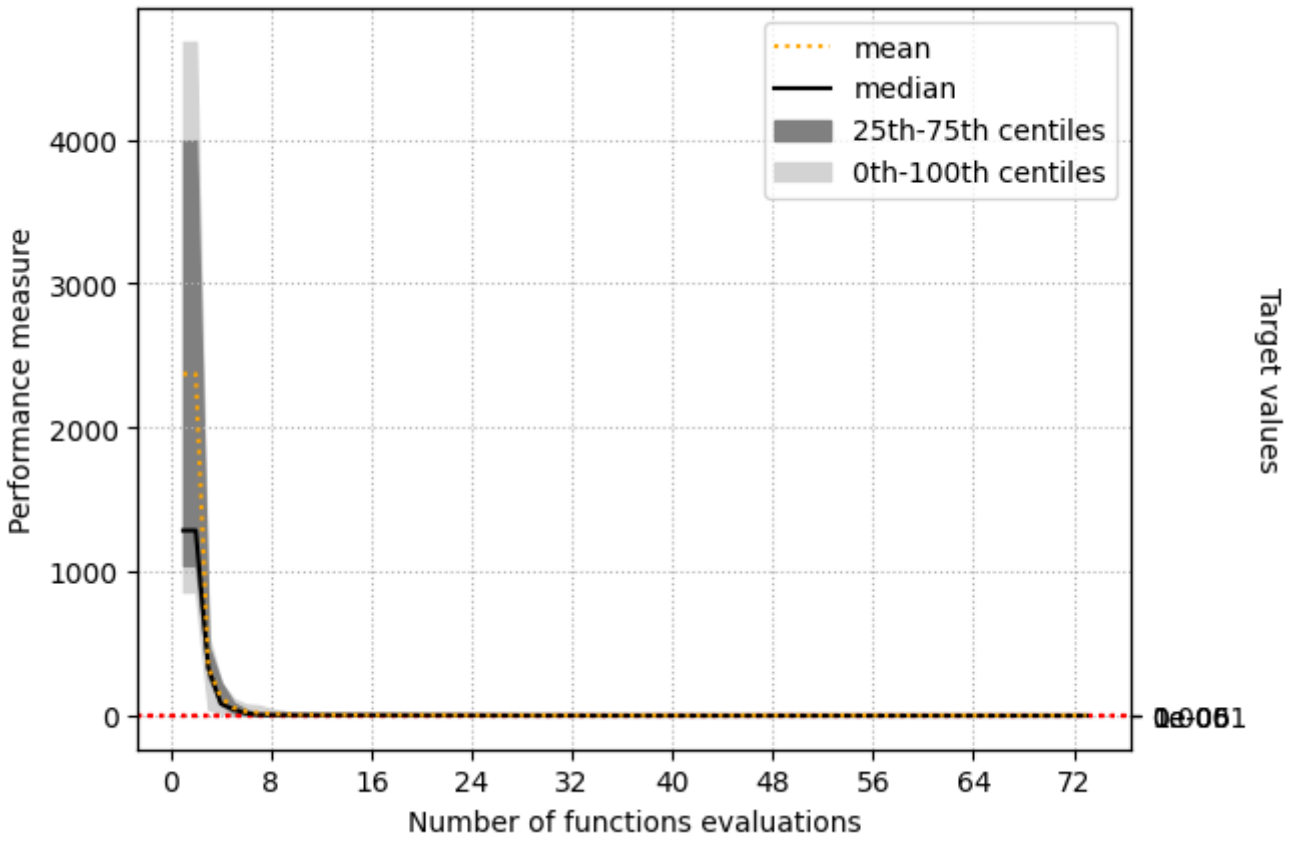
The performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem 'Rosenbrock 5D'.

	L-BFGS-B with 20 Hessian corrections
maximum	3.93084
75th centile	3.93084
median	5.67236e-15
25th centile	1.70659e-17
minimum	1.06233e-18

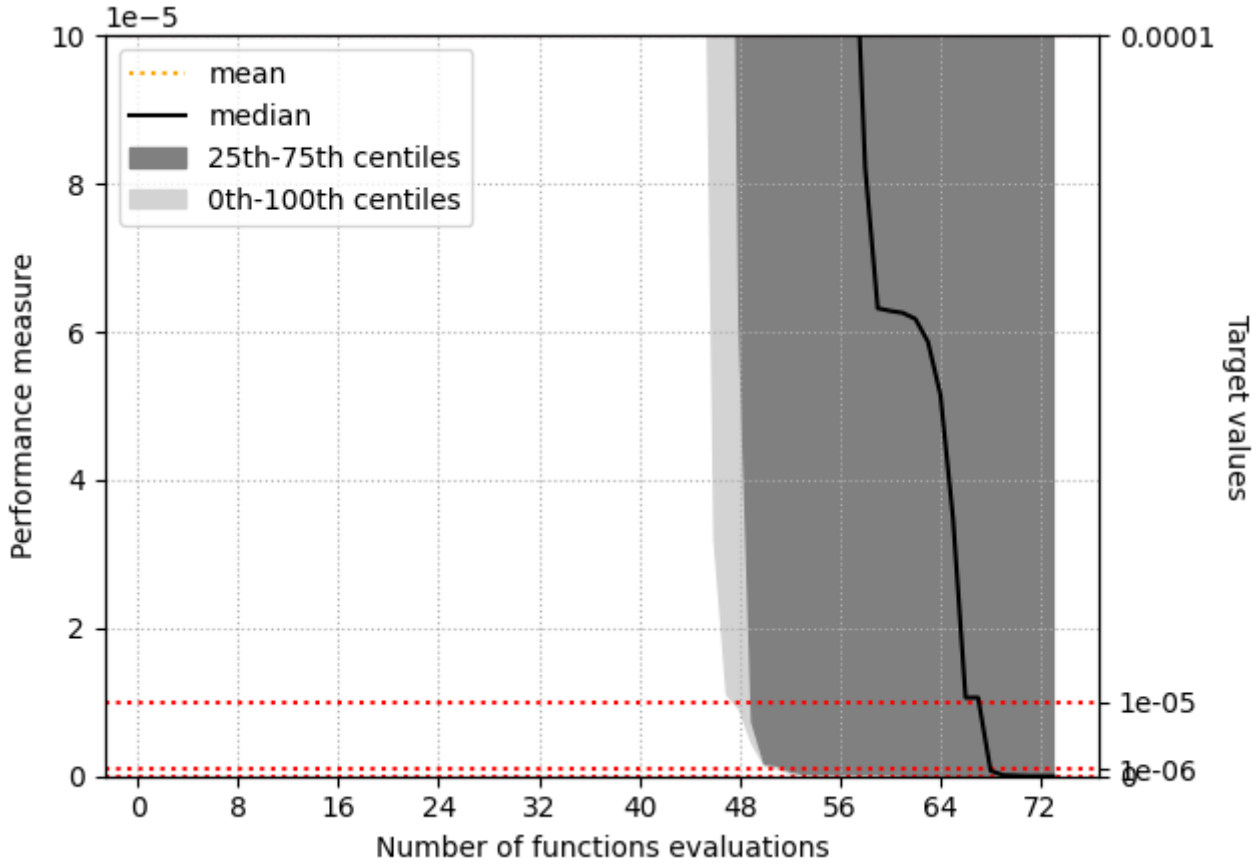
The final feasible performance measure of algorithm configuration 'L-BFGS-B with 20 Hessian corrections' for problem Rosenbrock 5D.

L-BFGS-B WITH 2 HESSIAN CORRECTIONS ON ROSENBROCK 5D

Performance measure



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rosenbrock 5D'.



The performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem 'Rosenbrock 5D'.

	L-BFGS-B with 2 Hessian corrections
maximum	3.93084
75th centile	3.93084
median	1.5986e-10
25th centile	1.2383e-10
minimum	2.30683e-11

The final feasible performance measure of algorithm configuration 'L-BFGS-B with 2 Hessian corrections' for problem Rosenbrock 5D.

6.4 SLSQP

6.4.1 2D problems

The results of the group of algorithms configurations "SLSQP" on the group of problems "2D problems".

The algorithms configurations

- SLSQP

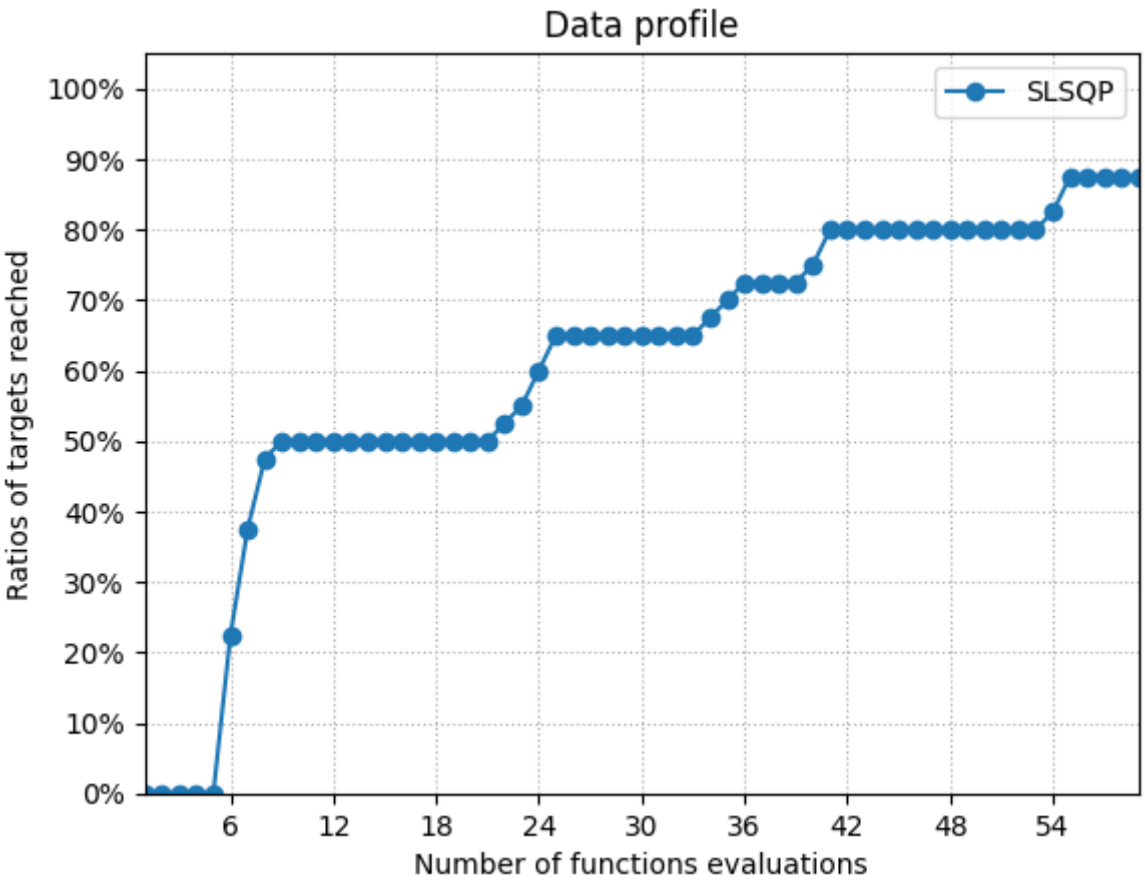
The problems

- Rastrigin
- Rosenbrock

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "2D problems" are represented in the following data profile.



The data profiles for group "2D problems".

RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

- Rastrigin
- Rosenbrock

6.4.2 5D problems

The results of the group of algorithms configurations "SLSQP" on the group of problems "5D problems".

The algorithms configurations

- SLSQP

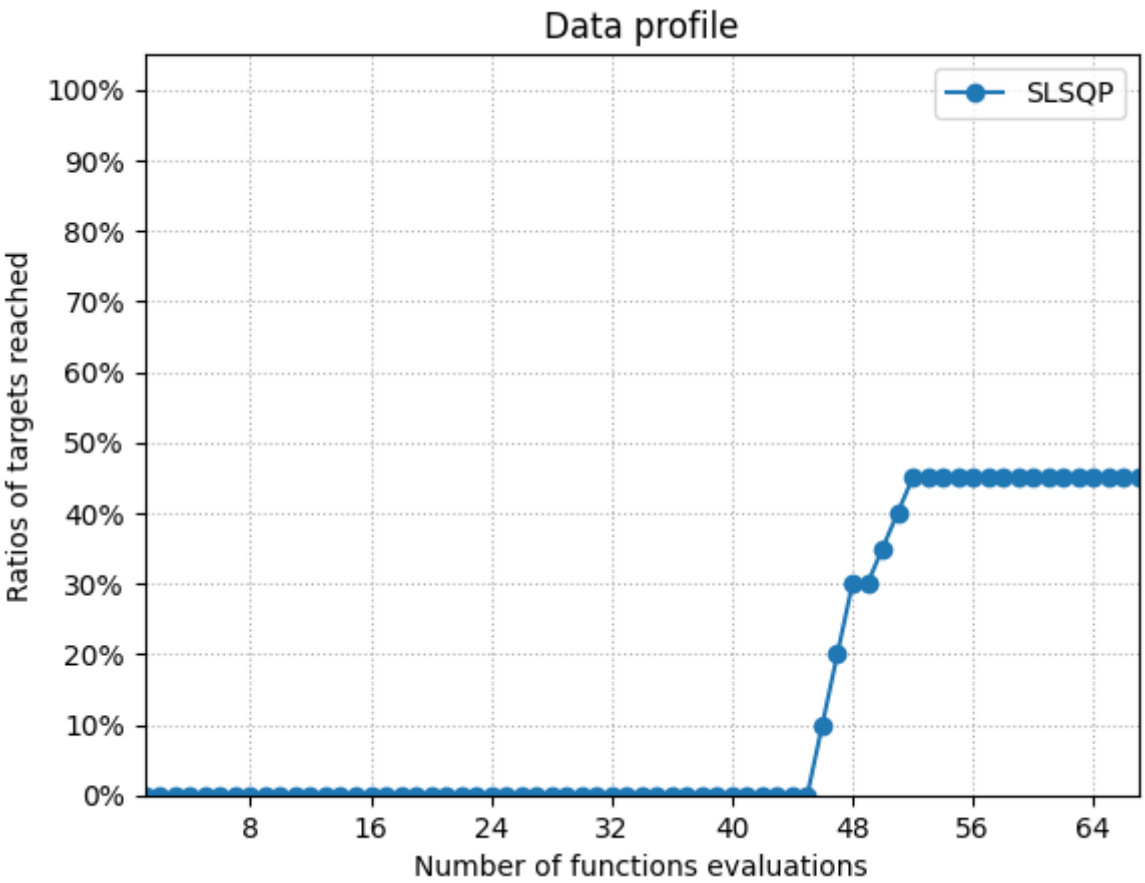
The problems

- Rosenbrock 5D

Benchmarking results

GLOBAL RESULTS

The performances of the algorithms on the reference problems of the group "5D problems" are represented in the following data profile.



The data profiles for group "5D problems".

RESULTS FOR EACH PROBLEM

The results of the algorithms configurations for each problem are linked below.

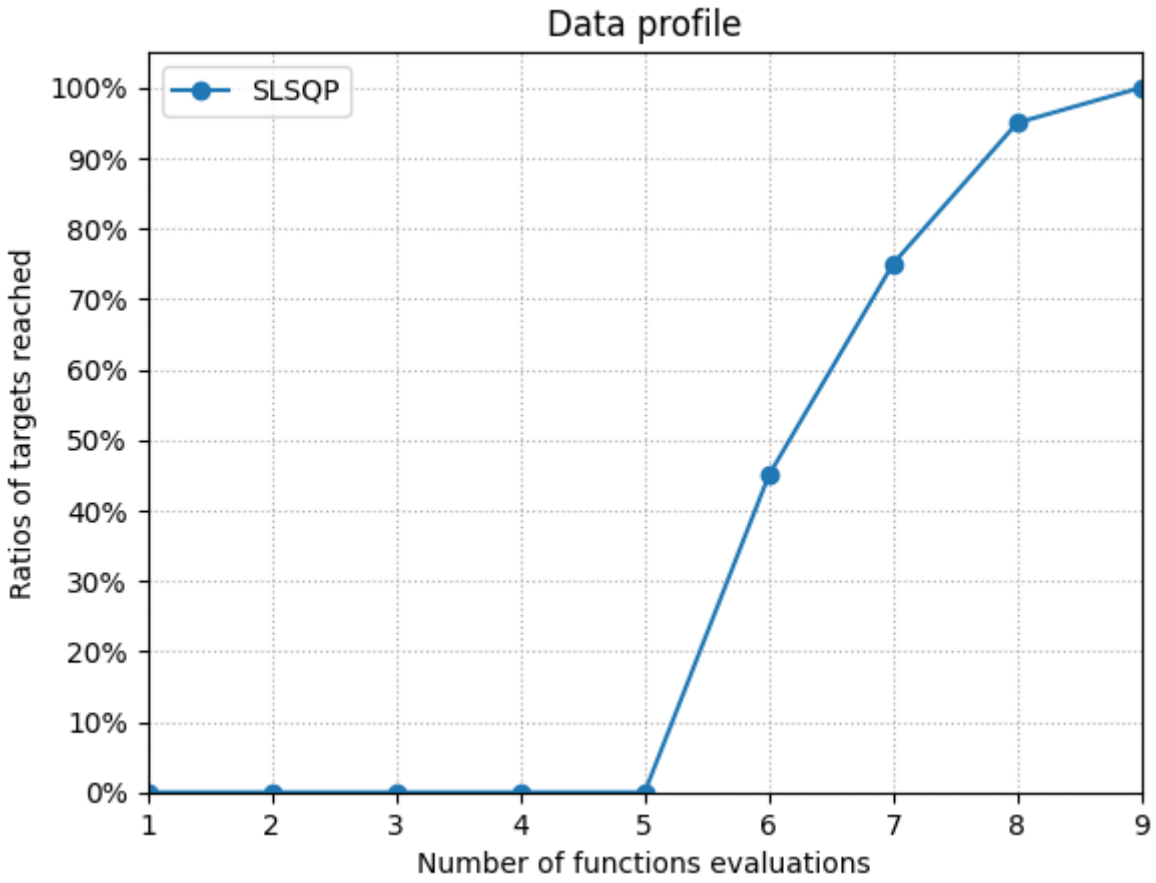
- Rosenbrock_5D

6.4.3 2D problems

Rastrigin

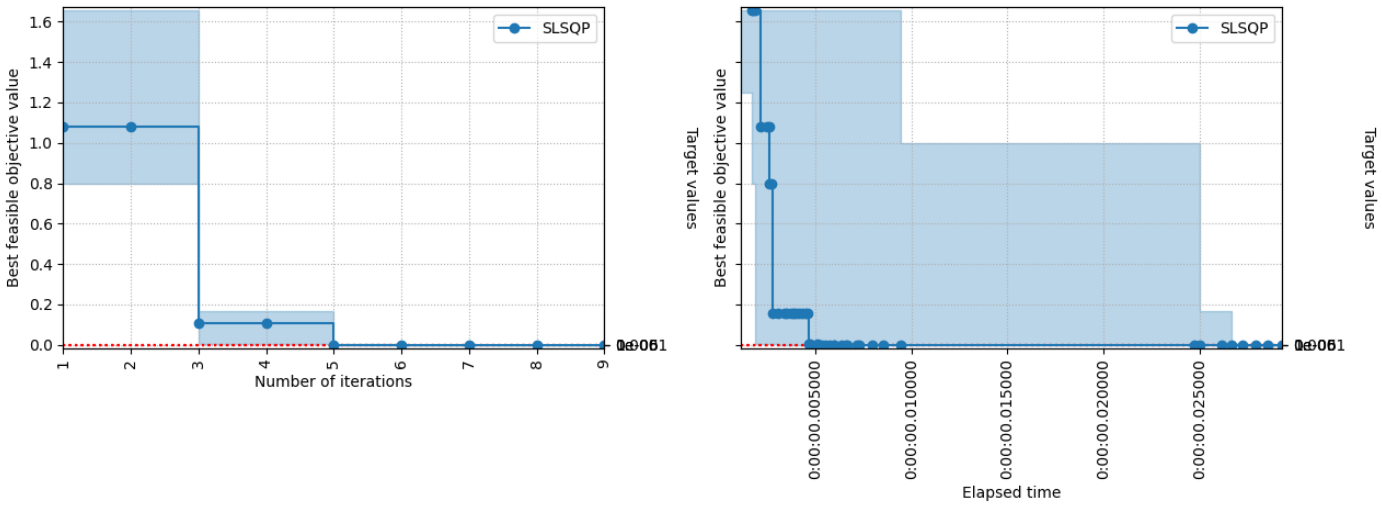
The results of the group of algorithms configurations 'SLSQP' for problem configuration 'Rastrigin'.

DATA PROFILES

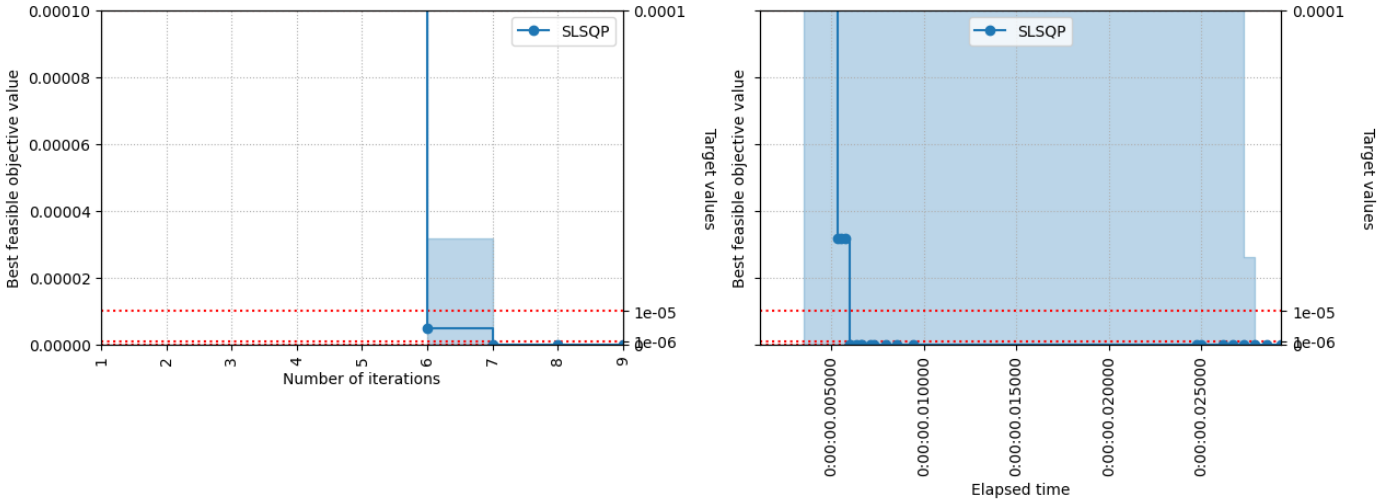


The data profiles of group SLSQP for problem configuration Rastrigin.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group SLSQP for problem configuration Rastrigin.

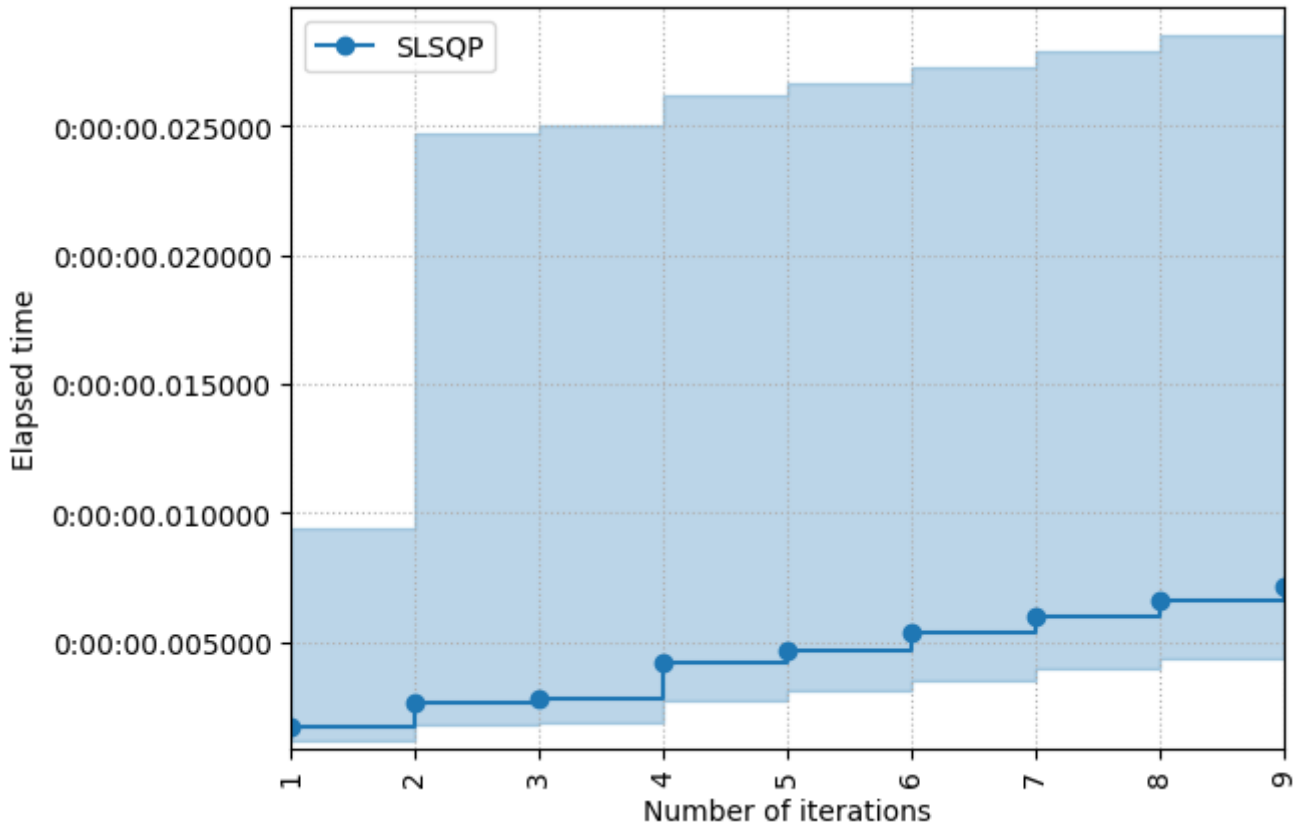


The best feasible objective value of group SLSQP for problem configuration Rastrigin.

	minimum	25th centile	median	75th centile	maximum
SLSQP	0	0	0	0	0

The final best feasible objective value of group SLSQP for problem configuration Rastrigin.

EXECUTION TIME



The execution time of group SLSQP for problem configuration Rastrigin.

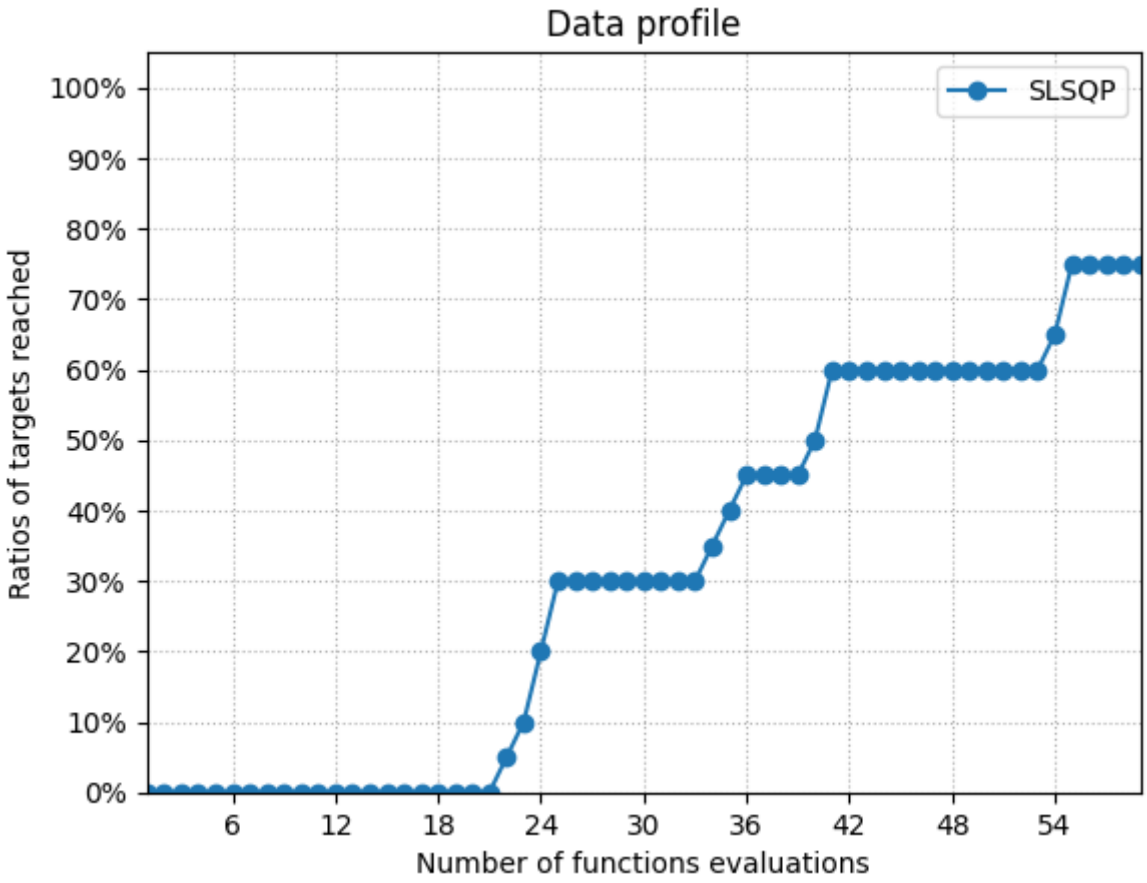
RESULTS FOR EACH ALGORITHM CONFIGURATION

- SLSQP

Rosenbrock

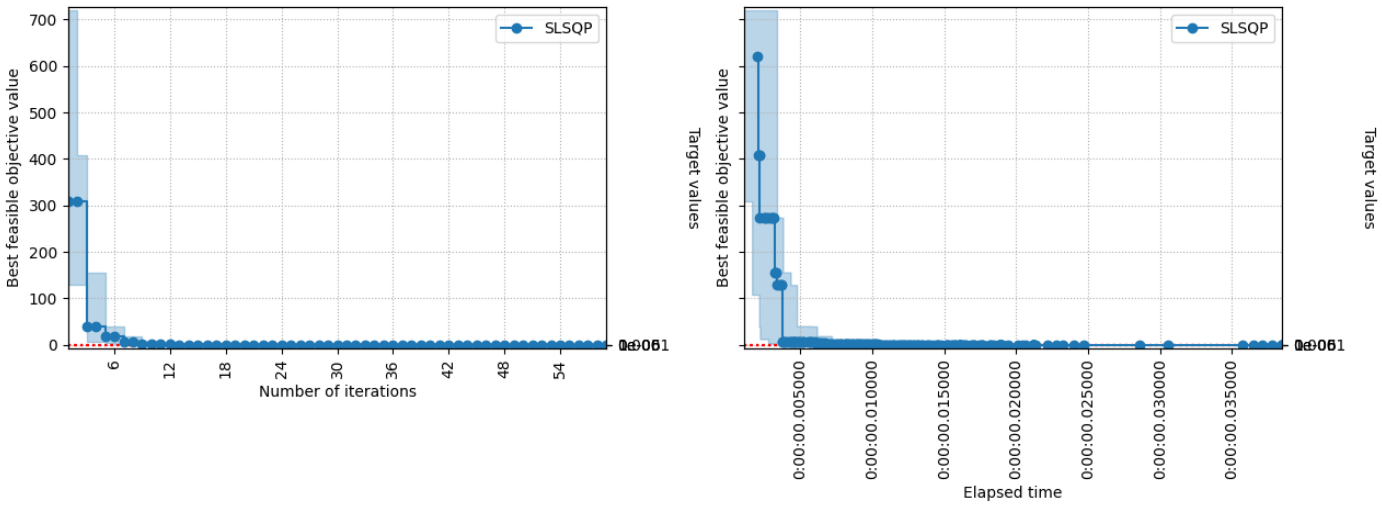
The results of the group of algorithms configurations 'SLSQP' for problem configuration 'Rosenbrock'.

DATA PROFILES

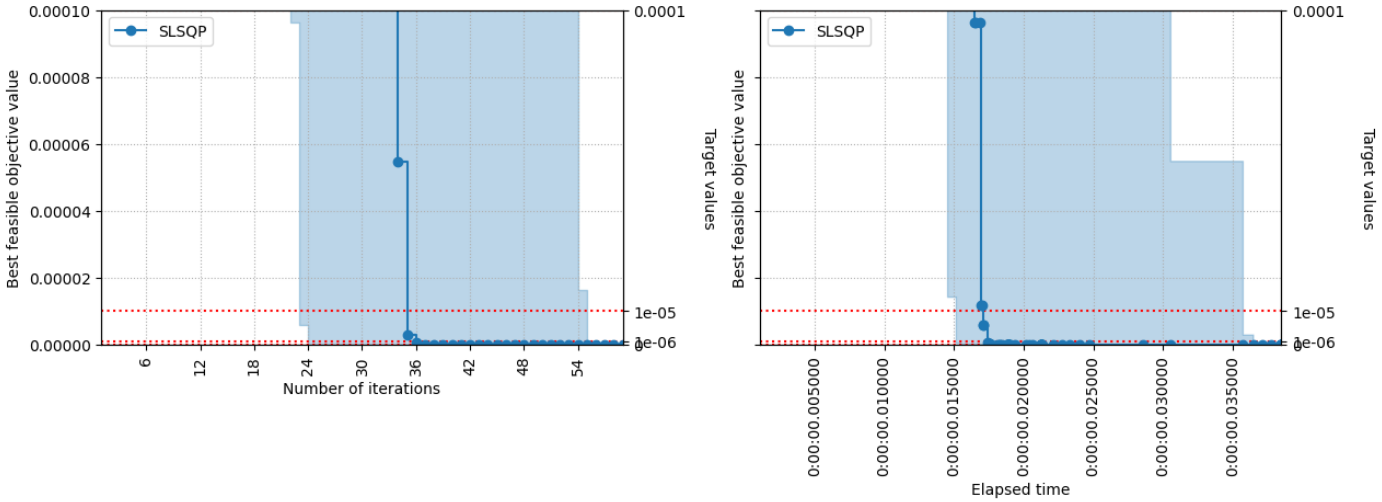


The data profiles of group SLSQP for problem configuration Rosenbrock.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group SLSQP for problem configuration Rosenbrock.

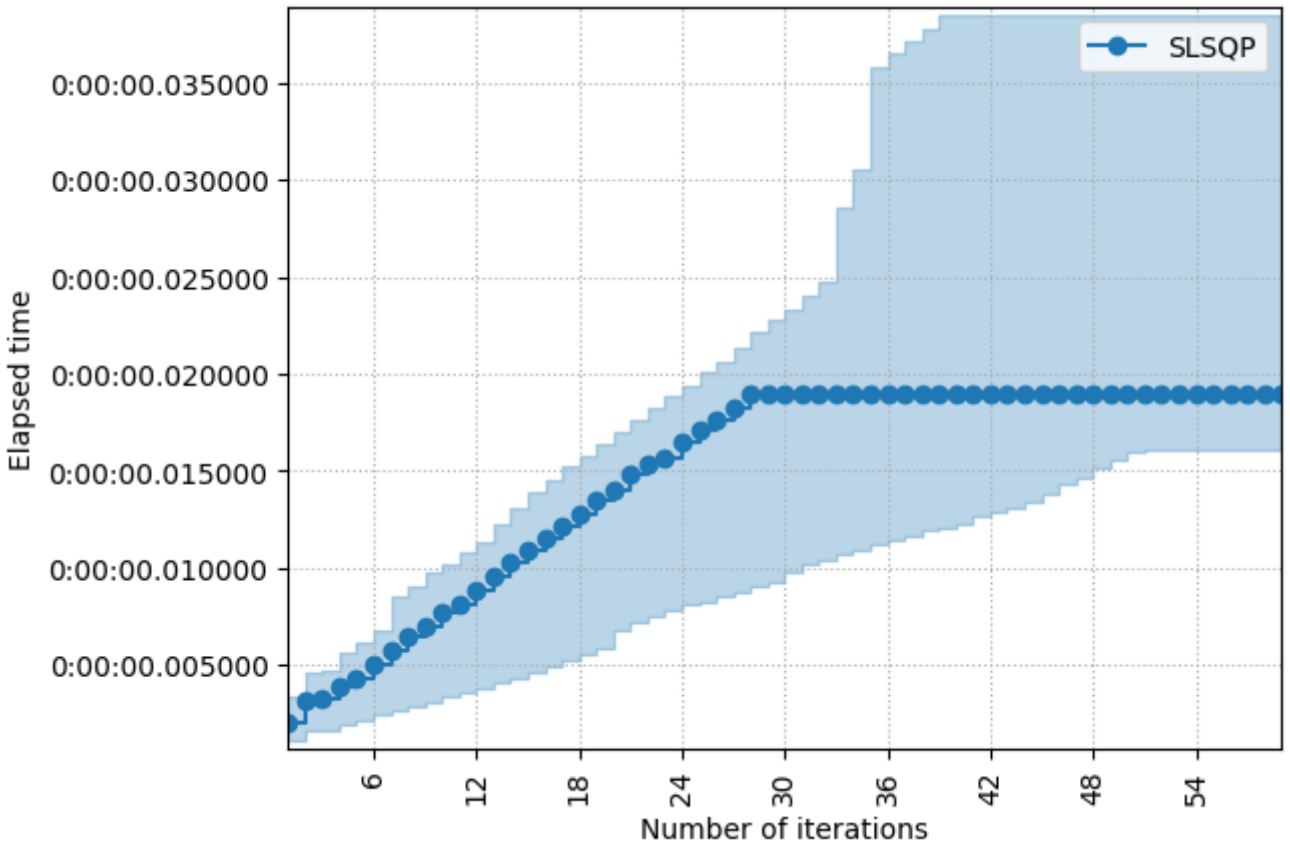


The best feasible objective value of group SLSQP for problem configuration Rosenbrock.

	minimum	25th centile	median	75th centile	maximum
SLSQP	1.18331e-20	4.23227e-20	2.01608e-17	3.74191e-17	5.45617e-14

The final best feasible objective value of group SLSQP for problem configuration Rosenbrock.

EXECUTION TIME



The execution time of group SLSQP for problem configuration Rosenbrock.

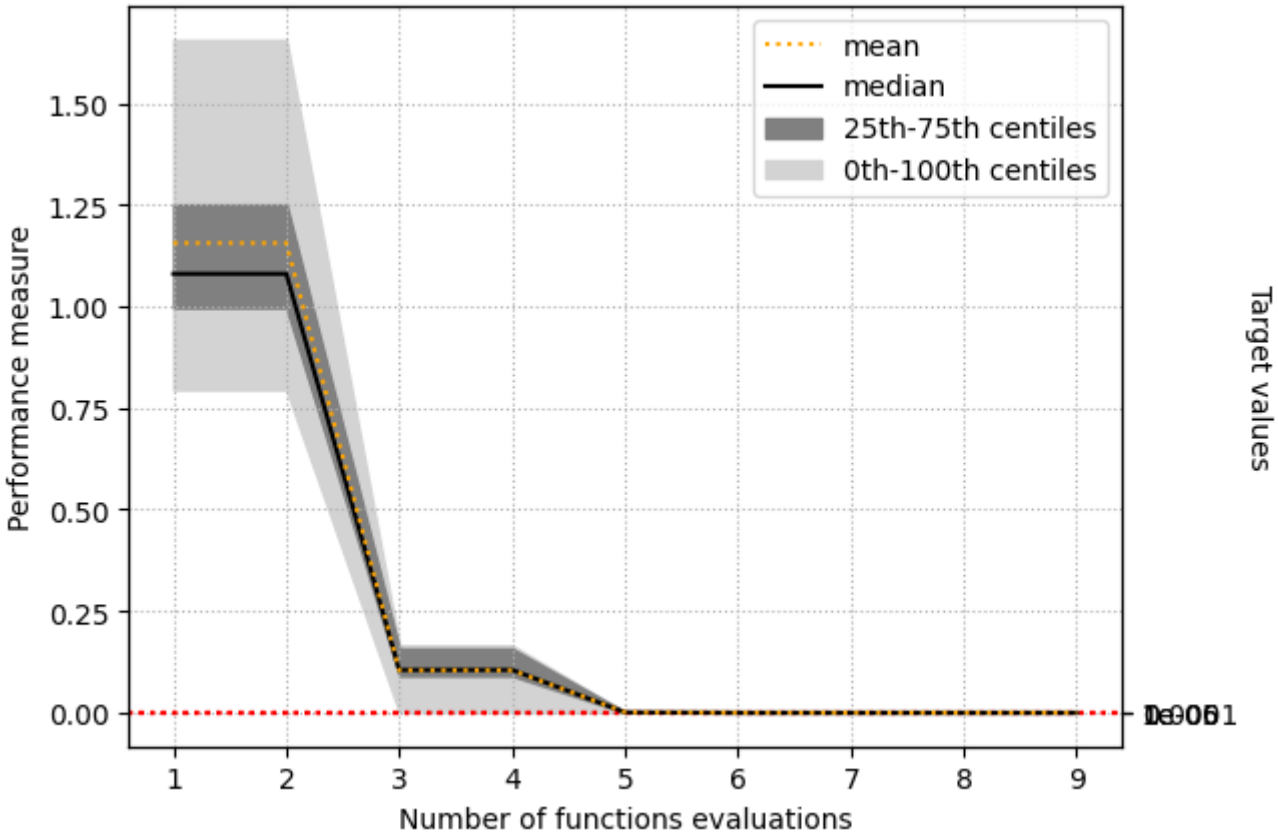
RESULTS FOR EACH ALGORITHM CONFIGURATION

- SLSQP

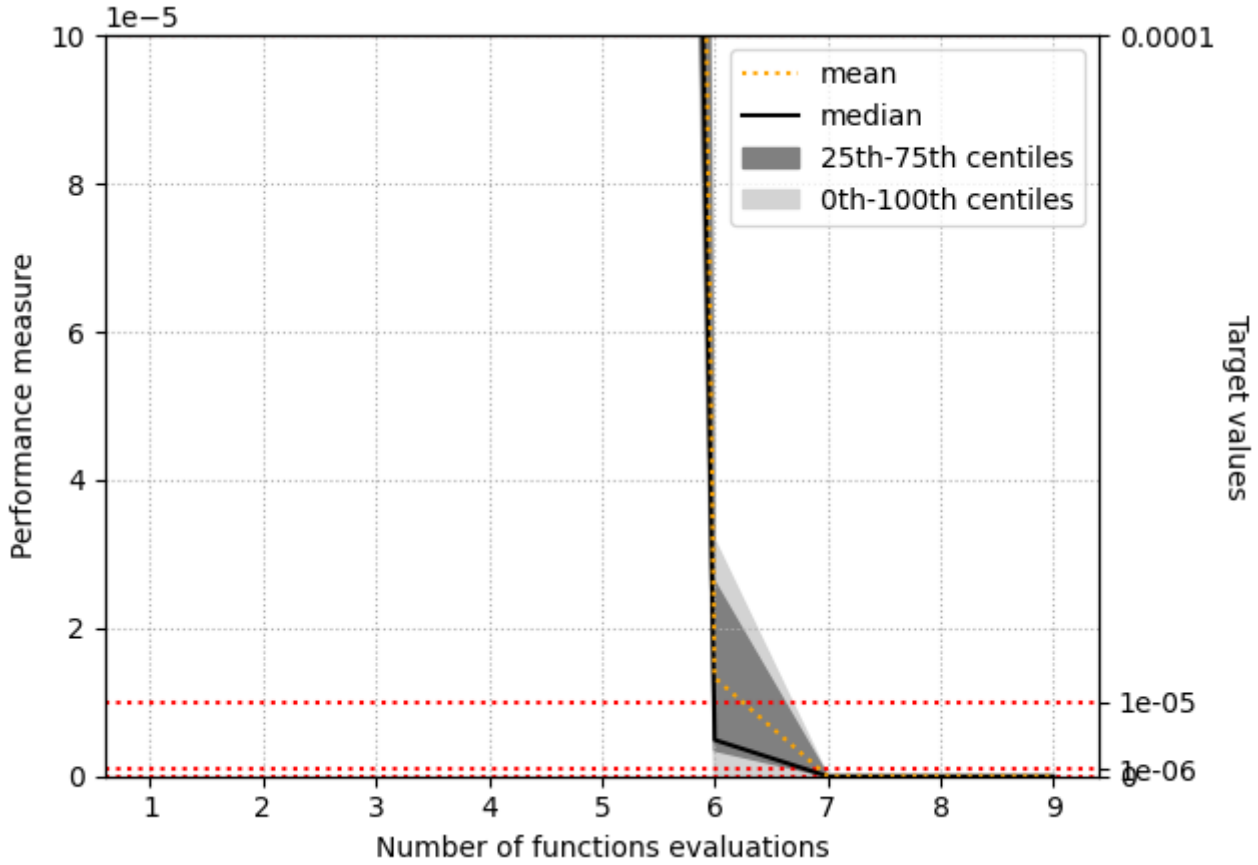
Rastrigin

SLSQP ON RASTRIGIN

Performance measure



The performance measure of algorithm configuration 'SLSQP' for problem 'Rastrigin'.



The performance measure of algorithm configuration 'SLSQP' for problem 'Rastrigin'.

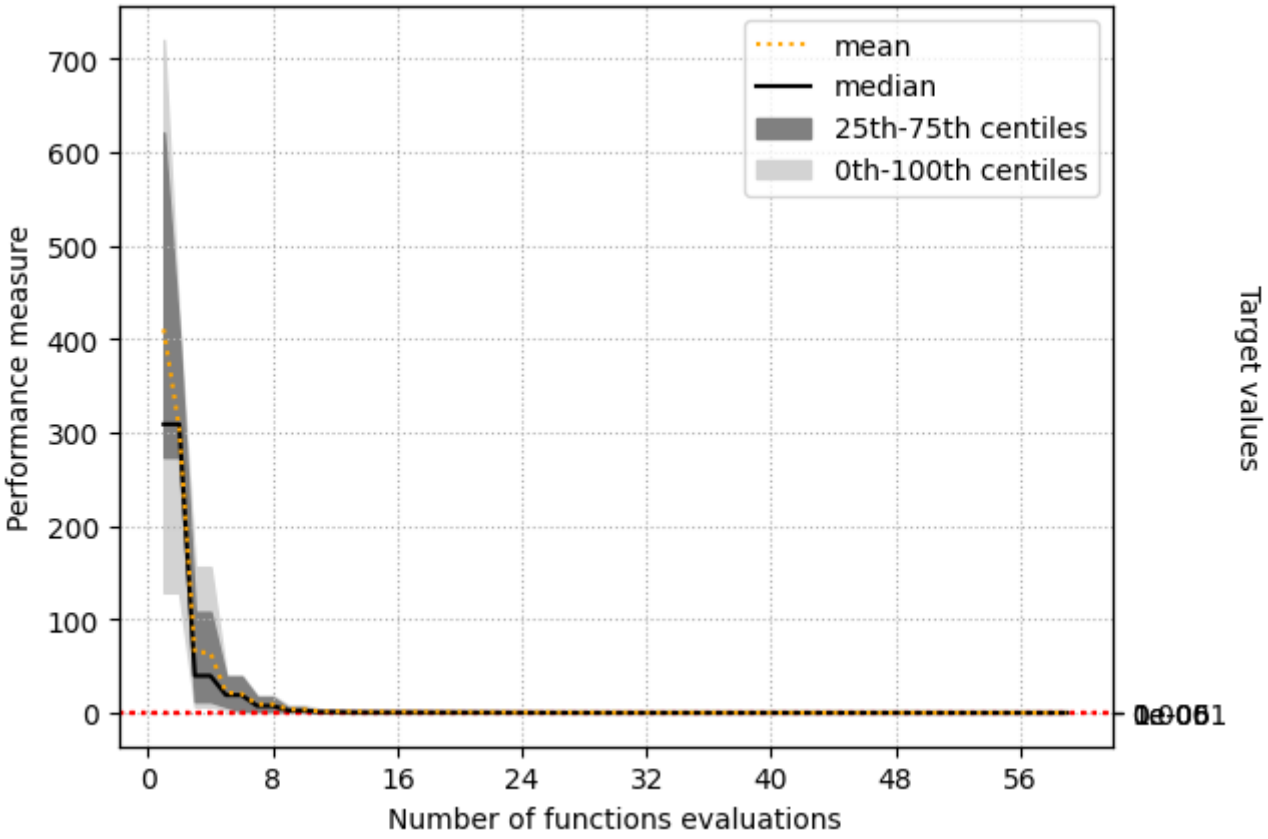
	SLSQP
maximum	0
75th centile	0
median	0
25th centile	0
minimum	0

The final feasible performance measure of algorithm configuration 'SLSQP' for problem Rastrigin.

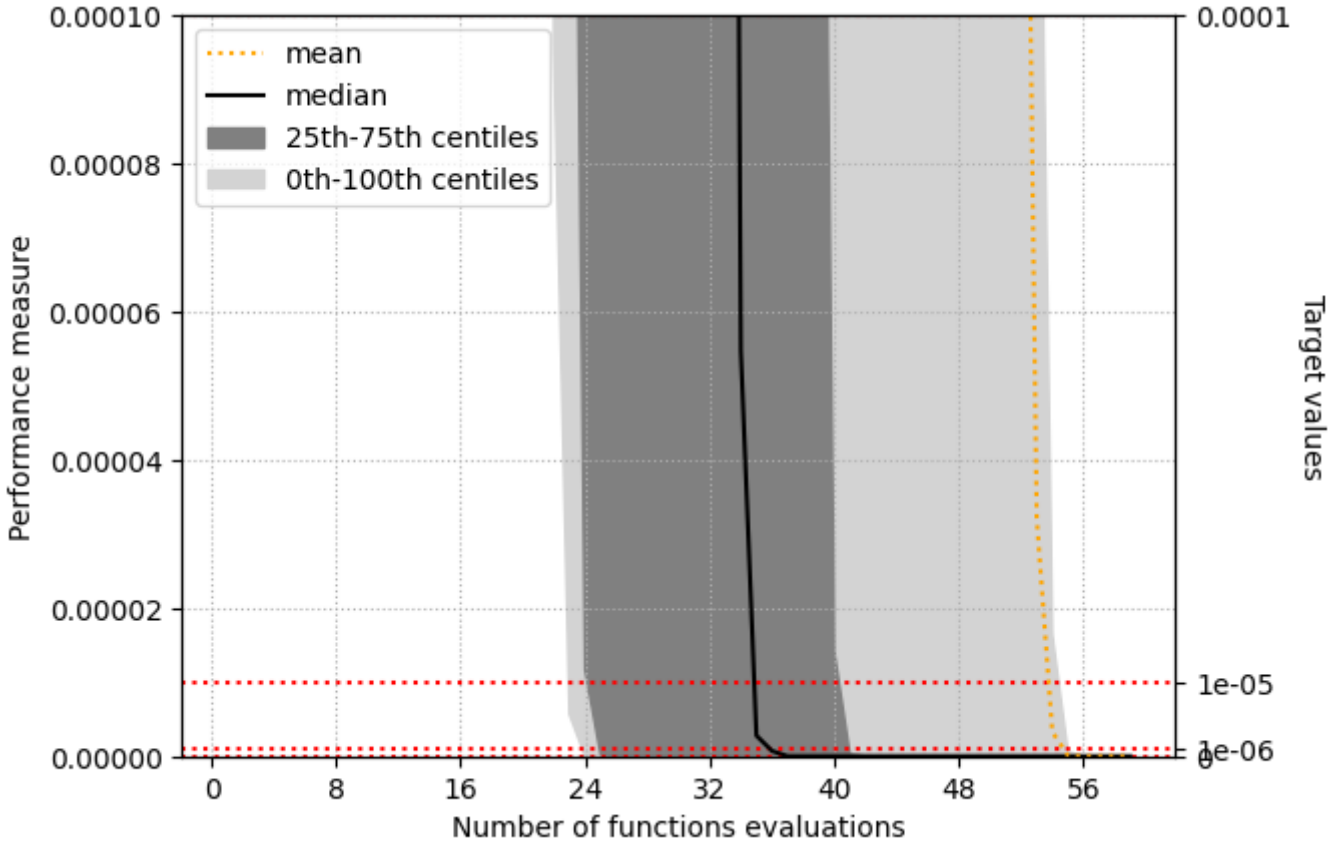
Rosenbrock

SLSQP ON ROSENBROCK

Performance measure



The performance measure of algorithm configuration 'SLSQP' for problem 'Rosenbrock'.



The performance measure of algorithm configuration 'SLSQP' for problem 'Rosenbrock'.

	SLSQP
maximum	5.45617e-14
75th centile	3.74191e-17
median	2.01608e-17
25th centile	4.23227e-20
minimum	1.18331e-20

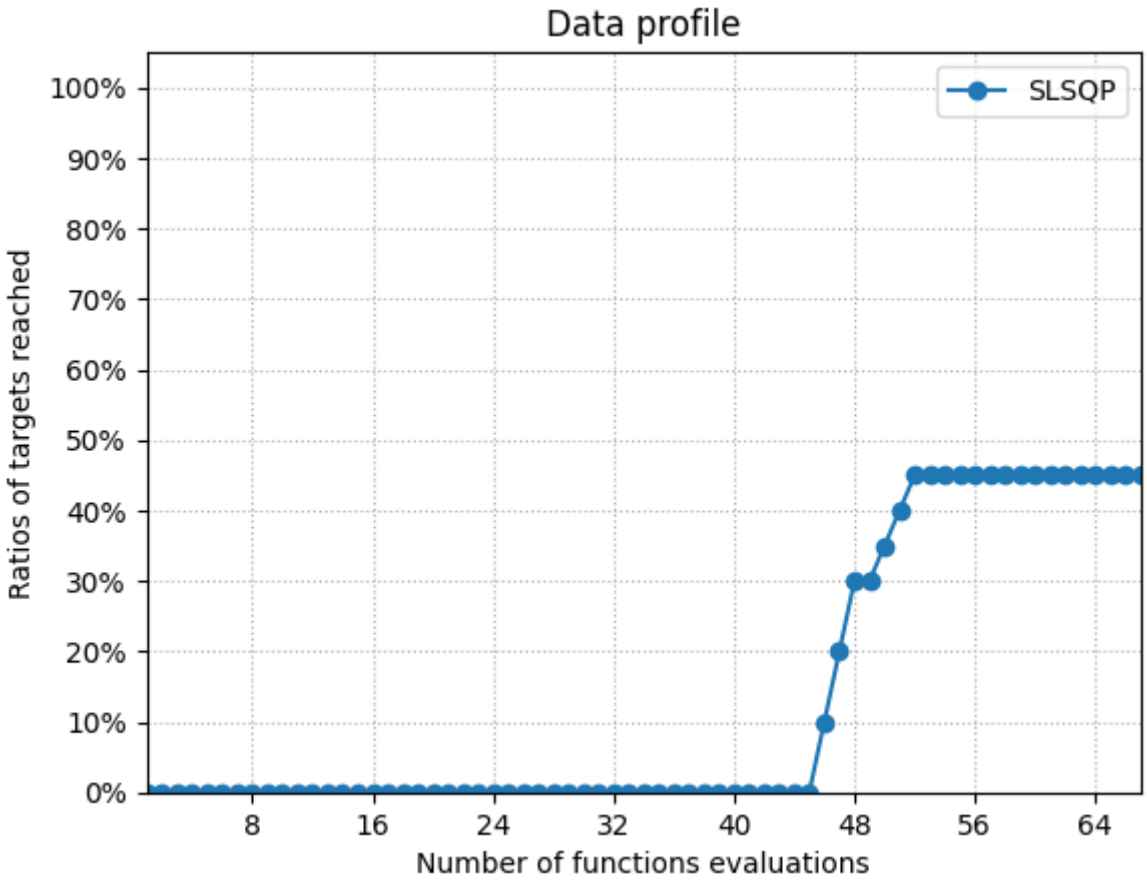
The final feasible performance measure of algorithm configuration 'SLSQP' for problem Rosenbrock.

6.4.4 5D problems

Rosenbrock 5D

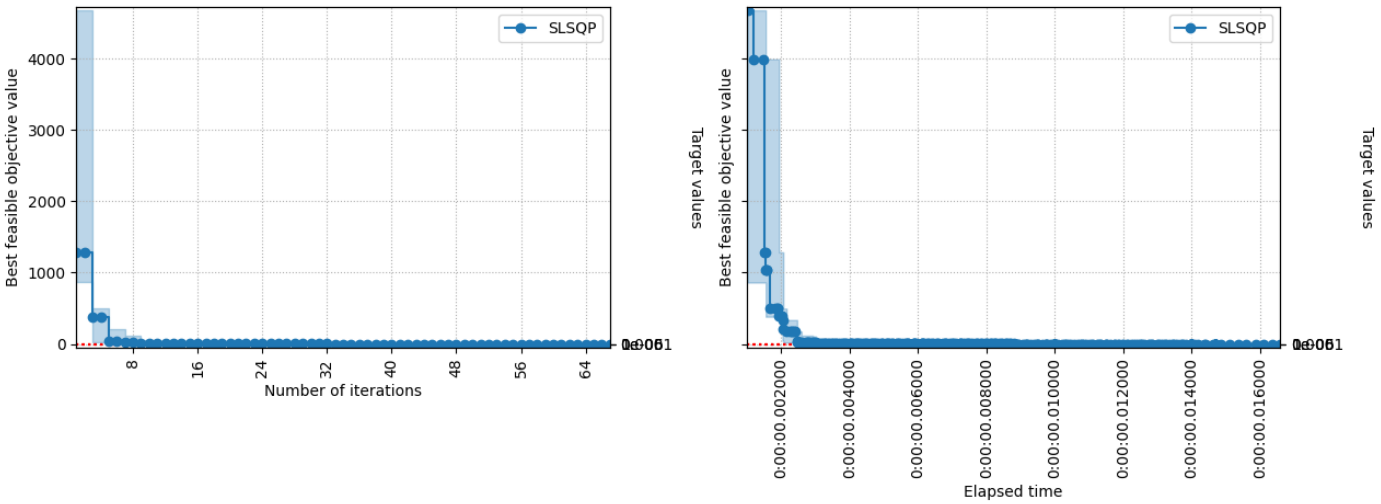
The results of the group of algorithms configurations 'SLSQP' for problem configuration 'Rosenbrock 5D'.

DATA PROFILES

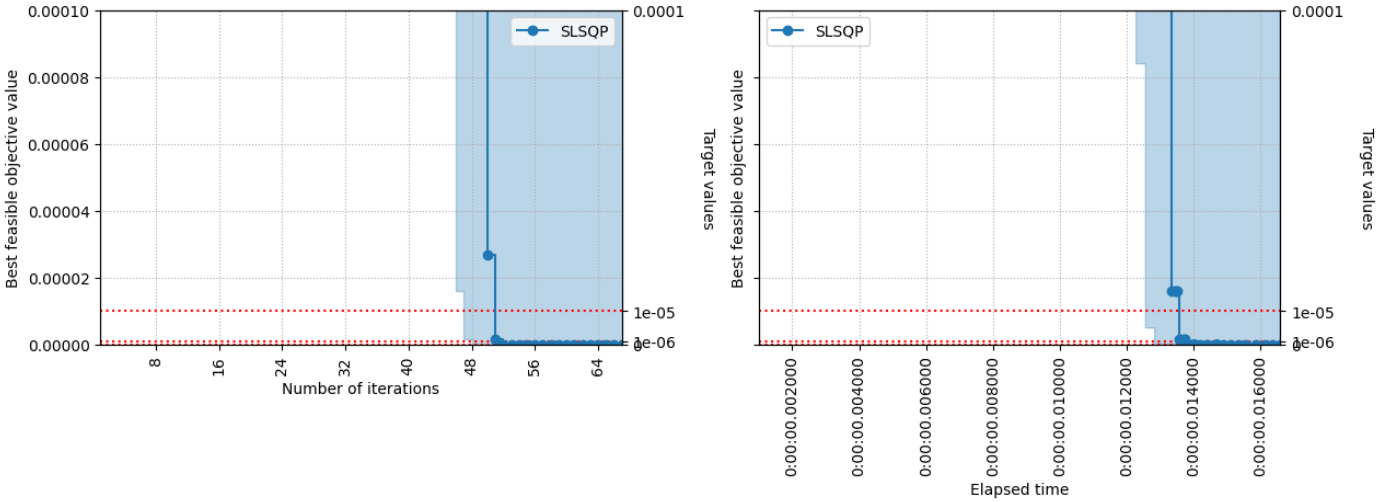


The data profiles of group SLSQP for problem configuration Rosenbrock 5D.

BEST FEASIBLE OBJECTIVE VALUE



The best feasible objective value of group SLSQP for problem configuration Rosenbrock 5D.

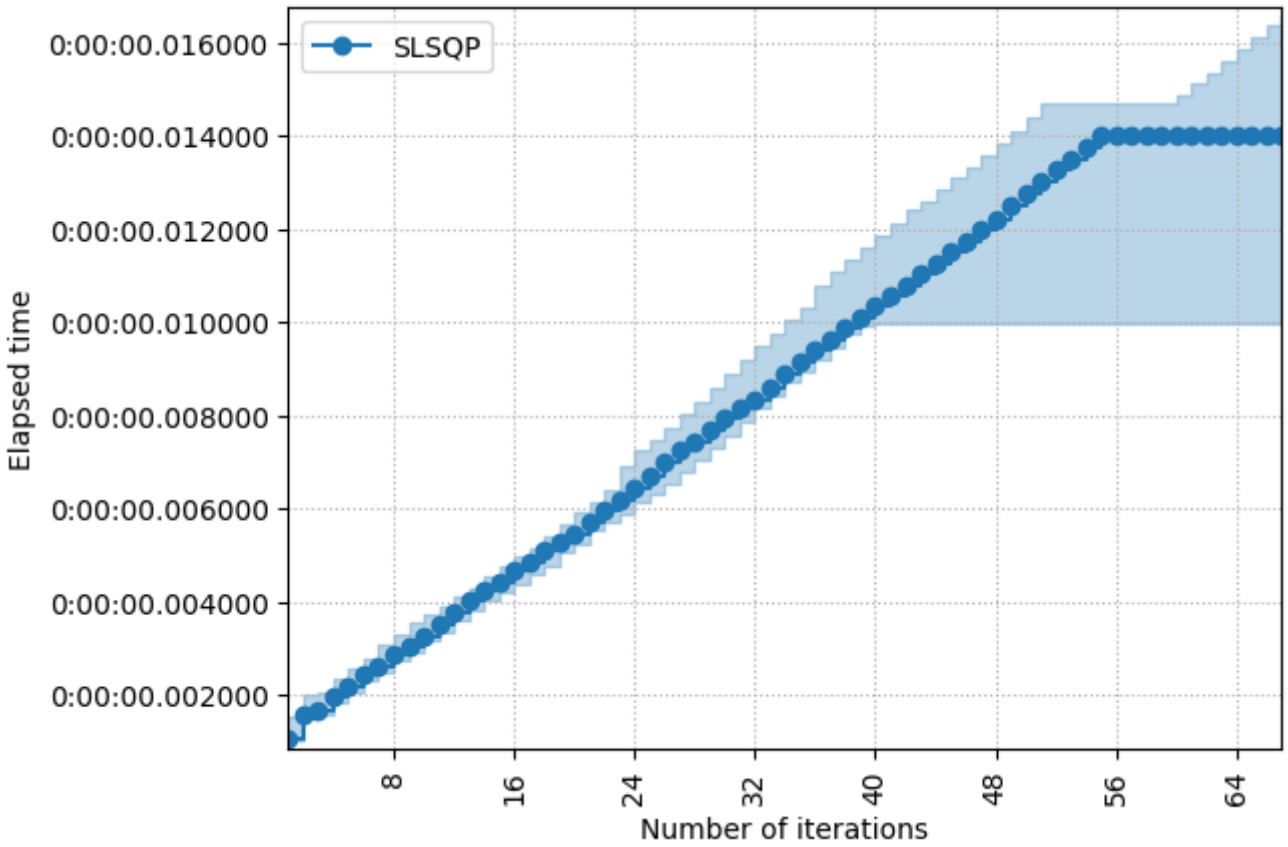


The best feasible objective value of group SLSQP for problem configuration Rosenbrock 5D.

	minimum	25th centile	median	75th centile	maximum
SLSQP	5.16019e-16	1.45728e-15	1.85869e-14	3.93084	3.93084

The final best feasible objective value of group SLSQP for problem configuration Rosenbrock 5D.

EXECUTION TIME



The execution time of group SLSQP for problem configuration Rosenbrock 5D.

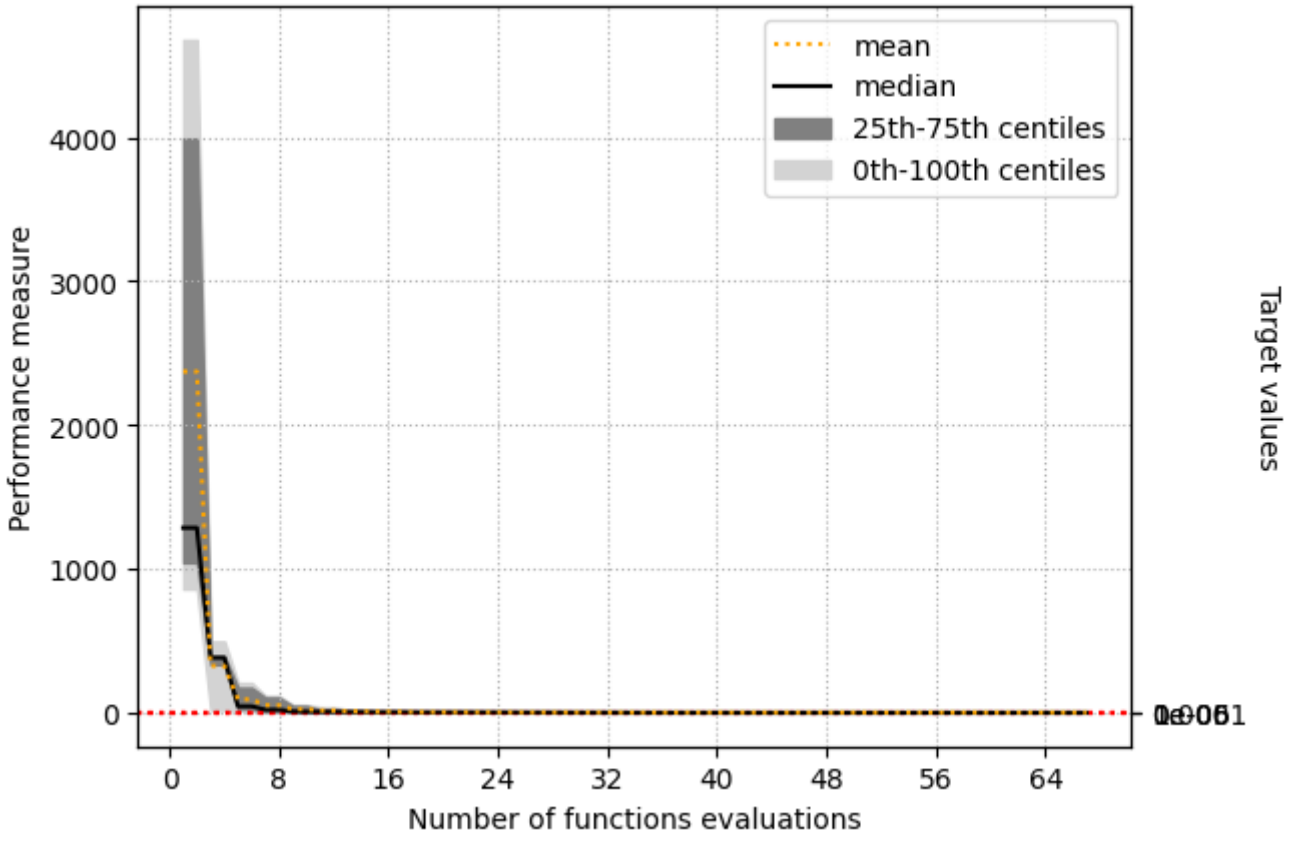
RESULTS FOR EACH ALGORITHM CONFIGURATION

- SLSQP

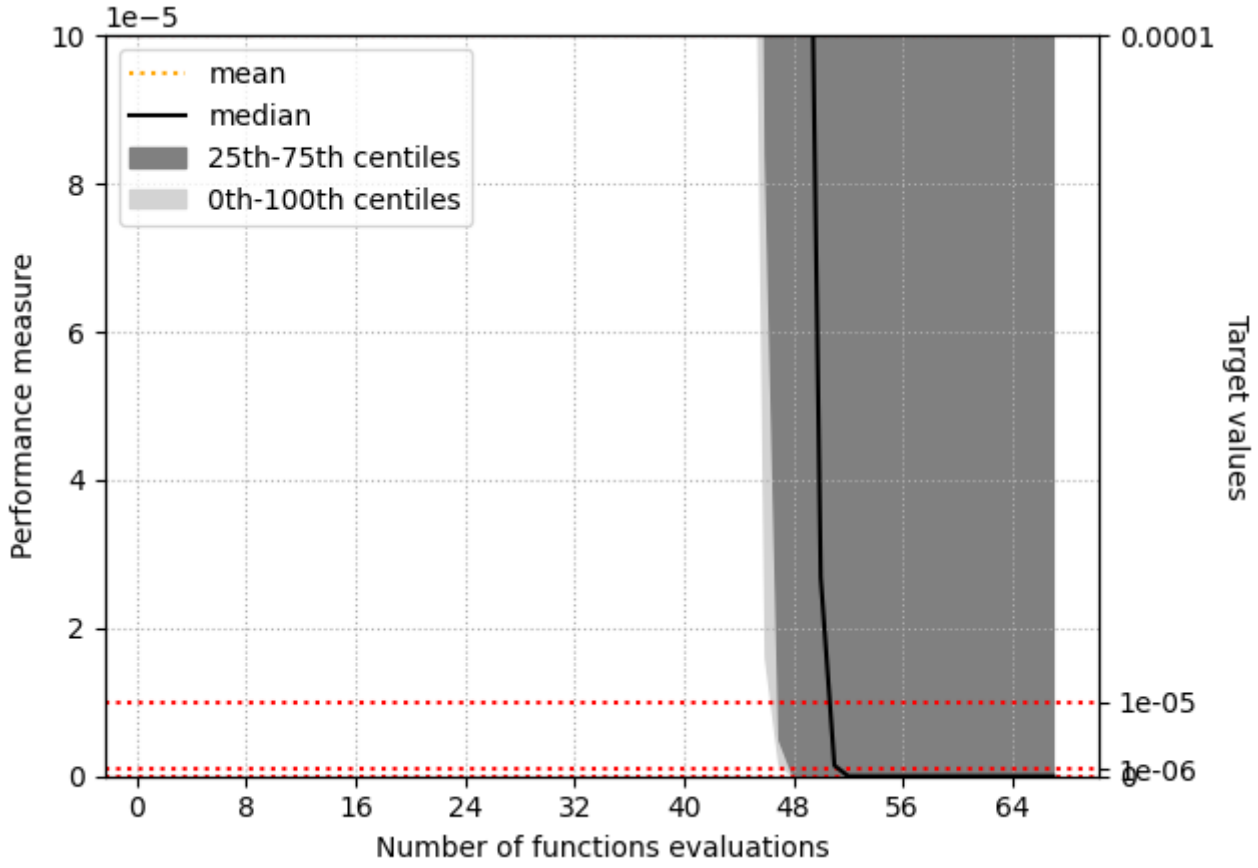
Rosenbrock 5D

SLSQP ON ROSENBROCK 5D

Performance measure



The performance measure of algorithm configuration 'SLSQP' for problem 'Rosenbrock 5D'.



The performance measure of algorithm configuration 'SLSQP' for problem 'Rosenbrock 5D'.

	SLSQP
maximum	3.93084
75th centile	3.93084
median	1.85869e-14
25th centile	1.45728e-15
minimum	5.16019e-16

The final feasible performance measure of algorithm configuration 'SLSQP' for problem Rosenbrock 5D.